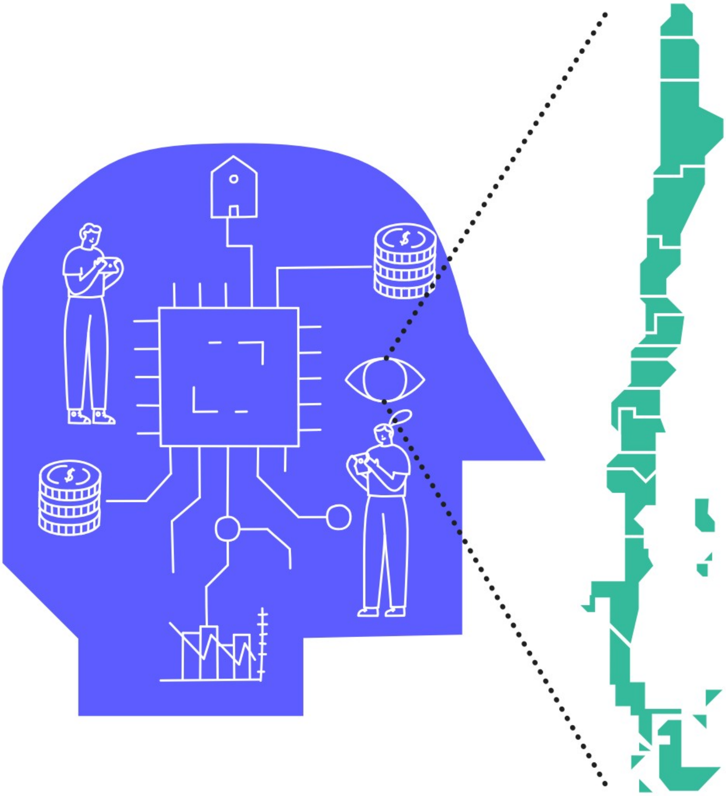
CHILE IA

POLICY NATIONAL OF **INTELLIGENCE ARTIFICIAL**

Ministerio de Ciencia, Tecnología, Conocimiento e Innovación

Gobierno de Chile



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SUMMARY EXECUTIVE

In October 2020, the Ministry of Science, Knowledge, Technology and Innovation (MinCiencia) published its first National Policy on Science, Knowledge, Technology and Innovation (CTCI). Its second axis, called "Future", proposes to "take advantage of knowledge, technology and innovation to anticipate, prioritize and build new and diverse forms of value, fundamentally anchored in challenges and singularities of the country", aiming to " build a society that looks responsibly and

"wisdom the future, understanding that, in any scenario, the CTCI play a vital role."

In compliance with the policy, while the MinCiencia was being installed in 2019, it was observed that Artificial Intelligence (AI), a general-purpose technology, would have a relevant impact on the future of the country, on its culture and its economy and that, compared to To this end, our country had to have a national strategy to anticipate its effects.

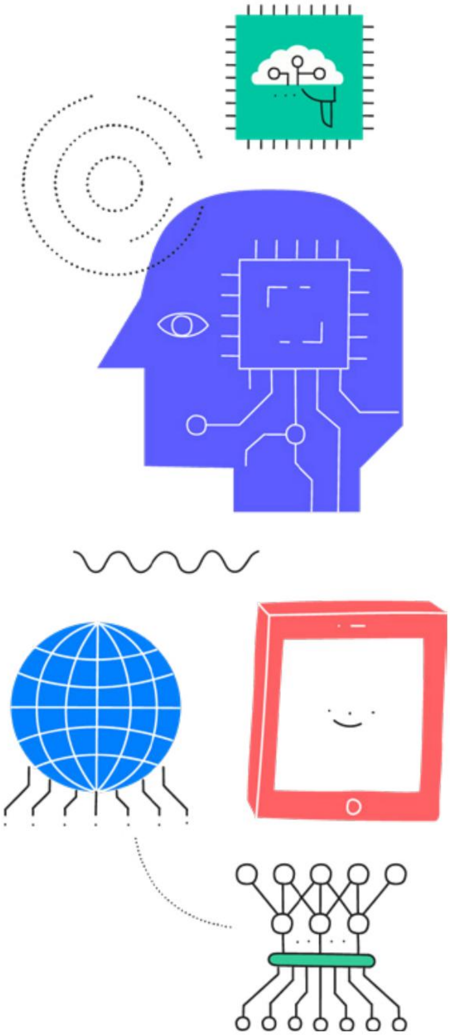
This policy emerges from undertaking that task collectively, and proposes, as a result of a participatory process, the objective of taking advantage of and promoting the country's capabilities to position ourselves above the OECD average and as the most advanced country in AI in Latin America and the Caribbean by 2031, inserted in the global vanguard, especially in training and attraction of talent, use of data, development and adoption, and in its appropriation by citizens in accordance with transversal principles of opportunity and responsibility.

During 2019 we presented the need to develop this policy to the President HE Sebastián Piñera, who commissioned this Ministry to prepare it, with special attention to ensuring that we have a specific action plan that derives from it, considering the unique opportunities of our country based on our history, our inhabitants. and our territory, and aiming to guarantee the future of Chile.

From then to date, an interdisciplinary and diverse group of 12 experts from academia, the productive sector and civil society was convened, and a public sector coordination process was promoted through an inter-ministerial committee.

Altogether, a national dialogue was enabled, calling in a broad and open manner to more than 1,300 people who participated.
in workshops, 400 people who participated in meetings in each of the regions of the country, and more than 5,300 people who attended 15 meetings in which AI was addressed from multiple perspectives and disciplines. All of the above was synthesized in a draft of this document that was

subjected to a public consultation process, in which more than 200 natural and legal persons participated. The input received during that process was systematized, analyzed in depth and consolidated together with ministerial discussions and with experts, resulting in Chile's first National Artificial Intelligence Policy.



Thus, this AI Policy ends up being based on four transversal principles: **AI centered on the well-being of people, respect for human rights and security; AI for sustainable development; Inclusive AI;** and **globalized AI.**

It is also structured in three axes: the first axis, **Enabling Factors,** refers to the structural elements that enable the existence and deployment of AI, such as the development of talents, technological infrastructure, and data; The second axis, **Development and Adoption,** includes the space where AI is created and deployed, that is, those who generate, provide and demand its different applications and techniques, which include the Academy, the State, the productive sector. and civil society; and the last axis, **Ethics, Regulatory Aspects, and Socioeconomic Impacts** addresses the new discussions that have arisen regarding human-machine interaction and the socio-technical system that it configures.

INTRODUCTION

WHAT IS ARTIFICIAL INTELLIGENCE?

A simplified and general definition developed by the University of Montreal understands Artificial Intelligence (AI) as "the set of computer techniques that allow a machine (for example, a computer, a telephone) to perform tasks that, common, require intelligence such as reasoning or learning."
(Dihlac et al., 2020, p. 4). The Organization for Economic Cooperation and Development (OECD) proposes a definition in the same line: "a computational system that can, for a given set of human-defined objectives, make predictions and recommendations or make decisions that influence real or virtual environments. AI systems are designed to operate with different levels of autonomy" (Organization for Economic Cooperation and Development, 2019, p. 12-13; Cabrol et al., 2020, p. 10).

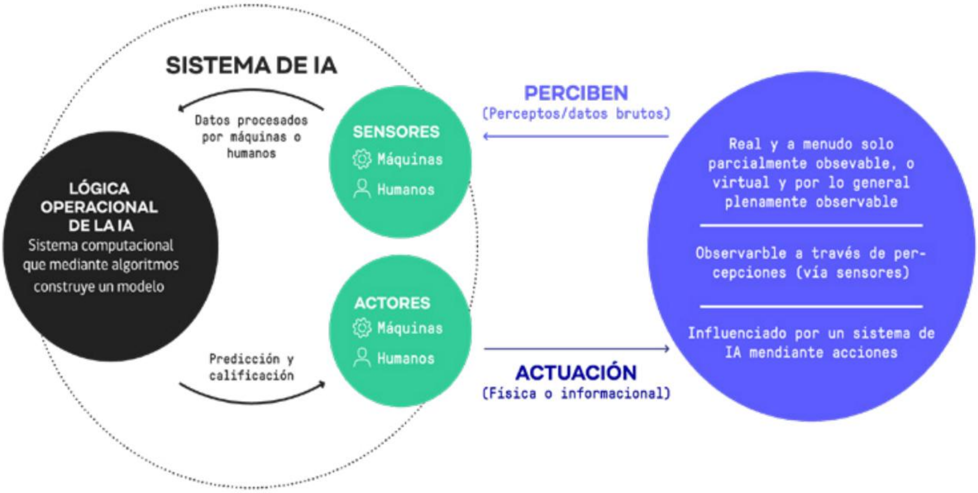


FIGURE 1: Conceptual Vision of AI. Sources: Gómez et al., 2020; Cabrol et al., 2020; Organization for Economic Cooperation and Development, 2019.

WHAT PLACE DOES AI OCCUPY IN OUR SOCIETY?

AI has acquired a leading role in recent years given its character as a general-purpose technology, the increase in productivity it provides and the abundance of factors that enable it (Bresnahan and Trajtenberg, 1995; Klinger et al., 2018).

Today it surrounds us, and is immersed in different sectors of the economy, knowledge and society. It recommends the best routes to reach our destinations, or the movies and songs that we might like, helps diagnose diseases in a timely manner, or analyzes large volumes. nes of astronomical data to find supernovae or planets like ours.

Although the concept began to be developed in the 1950s (Ronseblatt, 1950; Minsky, 1961), the adoption of this technology has accelerated in the last ten years thanks to the explosive increase in the amount of data available, the computing capacity, the number of people capable of developing it, and the advance of other complementary technologies, making possible and enhancing machine learning techniques.

WHAT WILL BE THE IMPACT OF AI IN OUR FUTURE?

AI is having a huge impact on the global economy and is likely to become even larger. For example, as a result of the development of AI, the consulting firm PricewaterhouseCoopers (PwC) estimates that the Gross Domestic Product (GDP) will be 14% higher by 2030 as a result of the accelerated development and adoption of AI (Rao et al., 2017). . The above, without considering the effect caused by the COVID-19 pandemic, which could further accelerate the digitalization and use of these tools at a global level (Chang, 2020).

The potential impact of AI, when analyzed in emerging countries, is dramatically reduced. In fact, the same PwC study estimates that Latin America and the Caribbean will barely receive 5.4% of the increase in world GDP (Rao et al., 2017). Along these lines, the region presents important challenges and exhibits less development than the leading countries, which is evident in a recent report from the IDB's fAIr LAC, a project that unites governments, universities and the private sector to promote ethical use of AI, which analyzes the twelve best-positioned countries in the region, finding gaps in digital infrastructure, education, gender gap in science, technology, engineering and mathematics (STEM). acronyms in English), patents, cybersecurity, decentralization, inequality, among others. Despite this, the same report highlights opportunities in the entrepreneurial ecosystem, the development of research in universities and research centers (Gómez et al., 2020).

Various AI applications are also transforming the services provided by governments, facilitating evaluations for social programs and creating new citizen service channels, complementing crime prevention actions with algorithms that detect fraud, contributing to mitigating risks in cities, or helping public health with systems that improve the accuracy of diagnoses or helping to prevent the spread of diseases, among many others.

Consequently, the adoption of AI by the State also represents an opportunity to improve the quality of life of citizens and face various challenges in emerging countries. However, Latin American States must make efforts to take advantage of this opportunity. The 2020 Government AI Readiness Index, developed by Oxford Insights and the International Re-search Development Centre, indicates that the best-positioned Latin American country is Uruguay, just in 42nd place, with Chile in second place, in 47th place.

The countries of Latin America and the Caribbean have also had a low participation in the discussion and elaboration of the res-

responsible, governance and ethical principles that should guide AI (Jobin et al., 2019; Fjeld et al., 2020). Chile, however, has been advancing in its positioning in the matter, and since 2020 it has participated in the Multisector AI Advisory Committee from the Digital Cooperation Agenda (Roadmap for Digital Cooperation) of the General Secretariat of the Organization of United Nations (UN). This committee seeks to raise projects and share experiences on how AI affects these five pillars: trust, human rights, security, sustainability and encouraging peace.

Given that in the region there is a level of development and discussion different from that of the northern hemisphere, it is urgent that emerging countries be present in international deliberative bodies—and that those who promote them make an effort -active effort to incorporate them—while advancing local principles and governance.

This has been recognized and addressed in the principles being developed by the United Nations Educational, Scientific and Cultural Organization (UNESCO) that were publicly and openly consulted in 2020 (UNESCO). United Nations Educational, Scientific and Cultural Organization, 2020).

WHY DOES CHILE NEED THIS AI POLICY?

The challenge of the technological revolution poses the need to accelerate our adaptation to the changes produced by the massification of technologies such as AI in society and, given that AI is a general-purpose technology and its impact on productivity is transversal, to the opportunities of the future, it is key to empower ourselves in their development and employment. In fact, the country's future economic growth depends on AI to the point that Chile's growth rate could be increased by this technology by 1 percentage point for every 3 points of growth by 2035 (IDB, 2018).

The National Innovation Council for Development (CNID) proposed five dimensions of action related to this in 2019.

topic: talents and employment, technological capital, social capital, modernization of the State and ethical and regulatory framework; also proposing five opportunities: Chile as a global data science hub, entrepreneurship and innovation ecosystem, inclusive technological revolution, digital state and talents for the 21st century (National Council of Innovation for Development, 2019).

Along these lines, during 2019, a group of ministries led by the Ministry of Science, Technology, Knowledge and Innovation (hereinafter MinScience) presented a diagnosis on the need to develop a National AI Policy. I an-

later was complemented by the Commission Challenges of the Future, Science, Technology and Innovation of the Senate of the Republic of Chile, which agreed on the need to generate a roadmap to promote research and development (R&D) in AI, innovation and thus capture its social and economic benefits (Commission on Future Challenges, Science, Technology and Innovation of the Senate of the Republic of Chile, 2019).

In addition, Chile adhered to the recommendations of the OECD AI Council, the first international standard on the matter.

This document proposes five principles: (1) inclusive growth, sustainable development and well-being; (2) human-centered and justice; (3) transparency and explainability; (4) robustness, security and protection; and (5) responsibility and accountability

accountability).

Furthermore, it proposes that countries adopt policies aimed at: (1) investing in R&D related to AI; (2) foster the digital ecosystem for AI; (3) shape a favorable and enabling policy environment for AI; (4) build human capabilities and prepare for the transformation of the labor market; and (5) cooperate internationally to have trustworthy AI (Organization for Economic Co-operation and Development, 2019b).

Finally, it is relevant to build a national path relevant to Chile regarding AI, which incorporates how Chileans will change with this technology in the coming years. The advancement of AI is conditioned and conditions, at the same time, the development of society.

Thus, the Policy that we present here covers both the technical specificity of AI, as well as its opportunities in our

economy, and the citizens' feelings regarding it, which contributes to a common imaginary or objective, which has been born from the encounter between the citizens of our country and the experts who contributed to this do- cumento, whose skills and experiences stand out in diverse fields such as AI, productivity, education, law, sociology, infrastructure, industry and public policies, among many others.

HOW WAS THIS POLICY PREPARED?

During the first half of 2019, Min-
Science together with the Ministries of Economy, Development and Tourism and the Secretariat General of the Presidency developed a comparative analysis of the AI strategies and policies published up to that period, observing the opportunity to be within the international vanguard of countries that have a national AI strategy, made up mostly of countries in the northern hemisphere.

Additionally, a review was carried out in the State in which more than 30 initiatives related to AI were detected in different public services that were in the process of design, implementation or operation.

During September 2019, the Ministry convened an interdisciplinary and diverse group of 12 experts to form part of a committee in charge of supporting the development of the base document of the Policy. During the same month, the Commission on Future Challenges, Science, Technology and Innovation of the Senate of the Republic of Chile (2019) presented to the President and the Minister of Science an elaborate diagnosis on the need for an AI strategy for Chile titled “Artificial Intelligence for Chile: The urgency of developing a Strategy.” As a result, two academics who led the writing of said document joined the Committee of Experts.

The Ministry of Science presented this preliminary analysis to President HE Sebastián Piñera, who commissioned the ministry to prepare a National AI Policy and an Action Plan. The main milestones of the process through which this Policy was prepared are summarized below.

In parallel with the formation of the Committee of Experts, an Interministerial Committee led by MinCiencia was created for the coordination of the ministries and services of the State that would participate in the task. The ministries that had begun to work in the diagnosis stage were joined by various

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its public services and new members as the development of the Policy progressed. Additionally, MinCiencia generated a network of counterparts in other ministries for review and to advance a registry of initiatives related to AI.

To incorporate active citizen participation, a process was designed that consisted of two stages. The first consisted of a broad and open call to contribute to the development of the Policy by publishing a tentative index of it, so that any person or organization could contribute with their vision, experience and knowledge. More than 1,300 people responded to this call. Next, tables were organized in all regions of the country, calling for 400 people who met to examine the topic considering the characteristics of their respective territories. In parallel, a series of 15 webinars were organized in which AI was addressed from multiple disciplines and perspectives and in which more than 6,600 people participated (available at bit.ly/Webinars-MinciencialA). The inputs generated during this process were systematized, analyzed in depth and consolidated together with ministerial discussions and with experts.

This analysis process together with dissemination made it possible to generate greater awareness about AI. The webinars and work tables were a space in which Chileans could co-construct around AI and, at the same time, learn about the technology, without being immersed in vertical instances of education. Figure 2 summarizes the scope of the first stage of the citizen participation process.

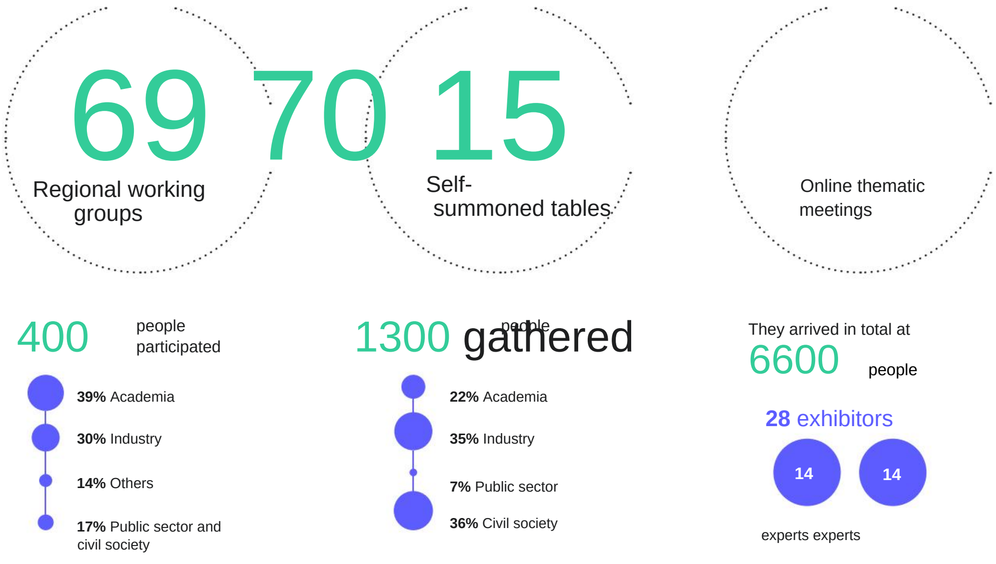


FIGURE 2: Synthesis of the National Artificial Intelligence Policy participation process.

The second stage was a public consultation on the first draft of the Policy. This consultation remained open between December 2020 and January 2021 and allowed 209 natural and legal persons to make detailed comments about it, which have been considered in this text. The characteristics of the participants and main results will be detailed in a complementary document, coming from the Public Consultation process according to the standards defined by the Legal Division of MinCiencia. Therefore, this document has been constructed considering the input received through the expert groups, the inter-ministerial team, the self-convened tables, regional tables and public consultation.

MinCiencia has the great challenge of positioning science, technology, knowledge and innovation as fundamental agents in the sustainable development of Chile.

The development of the AI Policy, as part of the broader Action Plan of the first Science, Technology, Knowledge and Innovation Policy generated under the country's new scientific institutions, points in the direction of empowering citizens and citizens in their relationship with science and technology, making the potential benefits their own, but also making us all responsible for mitigating their potential risks. This is how we move forward in charting our own path with a view to improving the quality of life of Chileans, and contributing to the development of our territories.

THEY RESPONDED

209

PEOPLE

SUMMARY

87,2%

AGREE AND STRONGLY AGREE WITH THE GENERAL OBJECTIVE SET FORTH IN THE DRAFT POLICY.

+80%

OF THE RESPONSES RECEIVED, THEY STATE THEY AGREE OR STRONGLY AGREE WITH THE PROPOSED AXES AND OBJECTIVES.

ON THE 3 AXES THEY INCORPORATED NEW VISIONS AND APPROACHES PROPOSED BY THE CONSULTATION PARTICIPANTS.

% AXLE **APPROVAL**

1. FACTORS ENABLERS

88,8% DEVELOPMENT OF TALENT

89,1% INFRASTRUCTURE TECHNOLOGICAL

87,3% DATA

2. DEVELOPMENT AND ADOPTION

88,8%

AGREE AND STRONGLY AGREE

3. ETHICS, REGULATORY ASPECTS AND SOCIOECONOMIC IMPACTS

83,3%

AGREE AND STRONGLY AGREE

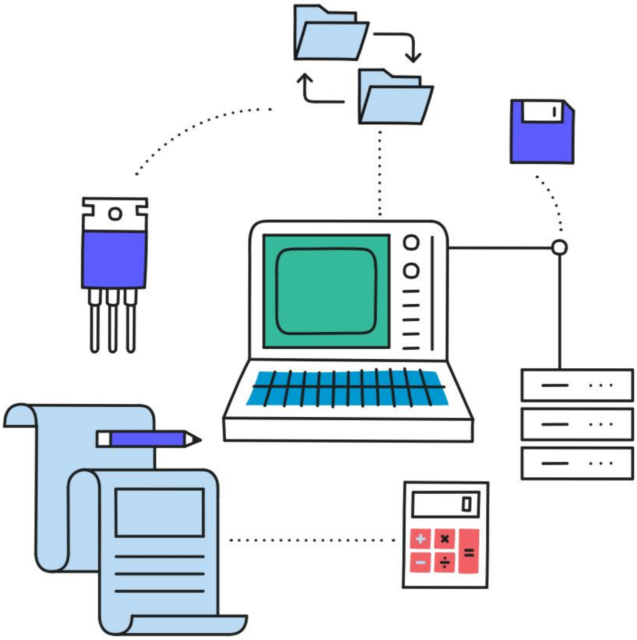
HOW IS THIS AI POLICY ORGANIZED?

This National AI Policy contains in the following pages an Objective, which operates as a conceptual pillar of the text, and which expresses the place in which we imagine the country driven by the initiatives that this policy organizes.

The text also contains four Principles, and is divided into three Axes, with interdependence, which correspond to Enabling Factors, Development and Adoption, and Ethics, Legal and Regulatory Aspects, and Socioeconomic Impacts. Each axis addresses the opportunities and gaps in its area, and introduces ce the objectives and priority actions that the country must undertake in a time horizon of 10 years to meet the objective of this policy in 2031.

Finally, the document has a glossary with terms that facilitate the understanding of technical aspects of the Policy.

This document is accompanied by an AI Action Plan, which specifies the initiatives included in each of the priority actions of the document, and includes those responsible and execution deadlines over the next decade.



AIM, BEGINNING AND AXLES POLITICS NATIONAL AI

AIM

Insert Chile into the vanguard and global collaboration related to AI, with an ecosystem of research, development and innovation in AI that creates new capabilities in the productive, academic and state sectors, and that is oriented according to transversal principles of opportunity. -ity and responsibility, contribute to sustainable development and improve our quality of life.

TRANSVERSAL PRINCIPLES

1.

AI FOCUSED ON THE WELL-BEING OF PEOPLE, RESPECT FOR HUMAN RIGHTS AND SECURITY

AI must contribute to well-being integral of people. The actions will be aimed at ensuring that people have a better quality of life, taking advantage of the benefits of AI and addressing its risks and potential negative impacts, with unrestricted respect for the human rights of the entire city. -

dadania. Both the algorithms and the data used to train the automated systems, especially

especially when it comes to data personal, they must be secure systems. The actions will contemplate evaluations of the risks and vulnerabilities in said processes in order to avoid the use of data sets that could lead to arbitrarily discriminatory decisions or results, and protect critical algorithms that, if violated, could compromise the security of our country.

2.

AI FOR SUSTAINABLE DEVELOPMENT

AI has great potential for emerging countries, like Chile, to diversify their economic matrix and make their industries more productive, as well as promote their ecosystem of research, technology, innovation and commercial applications derived from he.

The actions will be aimed at promoting the use and development of technology in the country, strengthening the ecosystem and incorporating AI as an axis of the country's sustainable development, that is, with social and environmental considerations. .

3. INCLUSIVE AI

Due to the close dependence that exists between the training of algorithms and the use of data for it, transparency and explainability become elements.

ments relevant to the conception of an inclusive AI, therefore, the actions will place special emphasis on the attributes of integrity and quality of the data to guarantee that its biases are known and treated appropriately.

AI should not discriminate based on protected categories or be used to the detriment of any group. In particular, it is of special importance that AI is developed with a gender and sexual diversity perspective, including historically relegated groups such as indigenous peoples, people with special abilities, or the most vulnerable sectors of our economy. , to become a useful instrument

for people, ensuring that gaps are reduced and closed.

Likewise, AI must be developed with special consideration for girls, boys and adolescents from a perspective of protection, provision and participation. This is justified on many occasions, this group does not have the opportunities or mechanisms

We must ensure their well-being and their incorporation when developing new technologies (Fund of the United Nations Children's Fund, 2020).

AI must be understood as a system that is changing the very nature of human beings and society. Along these lines, any action related to AI must address

occur in an interdisciplinary manner, enhancing the contribution of the various areas of knowledge. Furthermore, as a general-purpose technology, it is impossible to approach it from the exclusive perspective of experts. Knowledge and the construction of imaginaries around AI are distributed in society and, accordingly, they must constitute a system.

effort to integrate intelligence and collective feeling through open deliberation processes.

Finally, Chile has territories with diverse realities and characteristics throughout its geography. For this reason, the generation of regional and/or macro-zonal strategies will be promoted that allow taking advantage of the benefits of technology from the particularities of each territory and developing each region from its reality.

4.
GLOBALIZED AND EVOLUTIONING AI

The actions and initiatives with Regarding AI from Chile, they will consider how they fit into the context international, promoting participation in bilateral and multilateral spaces of which our country is a part. In addition, they will be aligned with the principles and agreements that Chile has signed, such as the OECD AI Principles, and will be updated in accordance with those that are signed in the future.

The global development of AI and the discussion on the ethical dilemmas it raises may conflict with local realities. For this reason, it will especially consider the unique characteristics of our country, taking into account our needs.

sities aligned with the evolution of technology and its enabling factors, which is why it will aim for its own path and in constant review, which is not necessarily the same as that followed by the countries of the northern hemisphere.

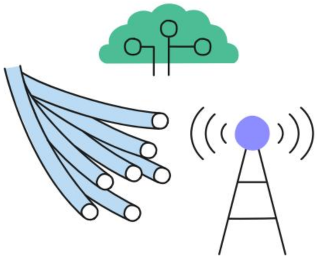
POLICY AXES

Based on the comparative analysis of the AI strategies published to date by various countries, and with the aim of facilitating their development and citizen deliberation, especially among groups with general interests or those that have a very specific focus, The AI Policy was structured into three axes. However, it is necessary to keep in mind that many of the topics discussed are interrelated, a connection that is represented in the principles indicated in the following section[1] .

1 Thus, the Policy recognizes a harmonizing role of its Transversal Principles. The numbering of the axes that are then stated does not necessarily mean prioritization or the establishment of an order of priority between them.

EJE Nº1.
ENABLING FACTORS

They are the basic elements that enable the existence and deployment of AI. That is, those components without which the use and development of this technology becomes impossible.



Included in this axis are the **development of talents, technological infrastructure,** and **data.**

EJE Nº2.
DEVELOPMENT AND ADOPTION

Understands the space where AI is developed and deployed. This space contains the actors that create, provide and demand its different applications and techniques, such as academia, the State, the private sector and civil society.

This axis includes **basic and applied research, technology transfer, innovation, entrepreneurship, improvement of public services, economic development based on technology,** among others.

EJE Nº3.

ETHICS, REGULATORY ASPECTS, AND SOCIOECONOMIC EFFECTS

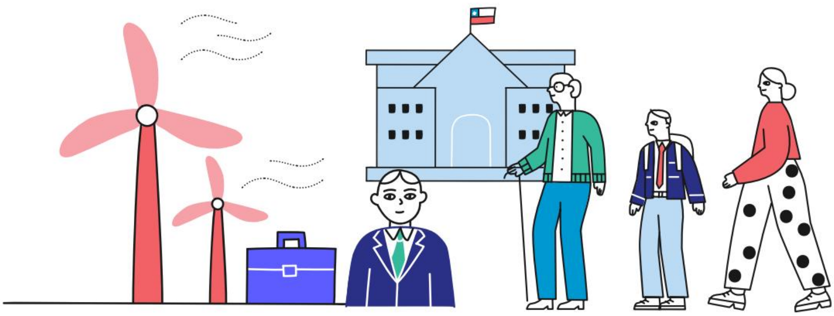
Being probably the most heterogeneous axis of Politics, it addresses the new discussions that have arisen regarding human-machine interaction and the socio-technical system that it configures, understanding that some of them are more advanced and others at an initial level of development.

Some of the most relevant regulatory aspects according to the discussion are included here.

citizen and national and international experience. For example, it addresses AI in consumer protection, privacy, the intellectual property system, and cyber security. In addition, it includes topics such as explainability/transparency, and gender and impact on the labor market among other ethical, social and economic aspects.

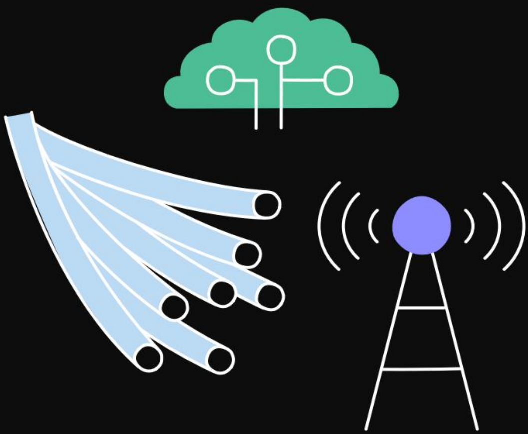
The following sections describe the topics addressed in each axis, the main specific objectives and the strategies proposed to achieve them. The details of the initiatives are included in the Action Plan that accompanies this Policy. Together, these initiatives will allow us to surpass the OECD average in various AI indicators related to

two with the three axes and advance to being the country that leads in the matter in Latin America and the Caribbean during the next decade.



AXIS 1.

FACTORS ENABLERS



1.1. TALENT DEVELOPMENT

The first requirement for both the use and development of AI is the presence of people with adequate training, experience and understanding in data, statistics, mathematics, engineering, signal processing, programming, among others. For this reason, we must direct efforts to provide, encourage and facilitate the training of talents in these subjects (OECD, 2019).

The current scenario in Chile is such that there are marked gaps in capacity and talents at all levels for areas related to digital transformation. For example, the Digital Talent program diagnosed a gap of approximately 65,000 professionals per year in digital technologies (Talento Digital, 2020).

The school level is an excellent moment. to include and prepare girls, boys and adolescents for a world in which the use of AI-based technologies is widespread, recognizing their influence and potential to be proactive users and citizens, while seeking to stimulate their interest in developing AI systems (Sandoval, 2018).

With regard to technical-professional training, there is a need to re-convert part of the workforce, especially people likely to be replaced by machines (Clapes UC, 2018). With respect to future workers, various studies have identified a gap between the skills provided by Higher Education Institutions (HEIs) and the competencies demanded by our industry, specifically in terms of skills related to Industry 4.0 (National Productivity Commission, 2018).

Regarding undergraduate and postgraduate university training, it is essential to have people dedicated to R&D in AI in Chile and in the world. At a global level, there is an increase in the shortage of people with these skills (OECD, 2019).

In our country, there is only one person dedicated to R&D for every thousand workers, 8 times below the OECD average (National Science, Technology, Knowledge and Innovation Policy, 2021).

Finally, given the nature of the evolution of AI, and the possibility of other technologies emerging with the potential to generate a relevant impact on its nature and deployment, we need to create and maintain anticipation exercises so that the development of talent continues to be aligned with the future of AI and other technologies yet to emerge.

To address these challenges, the following objectives are proposed:

OBJECTIVE 1.1.1

Promote the training of skills, knowledge and aptitudes for the use, development, understanding and analysis of AI in the school system, considering the positive and negative implications of the technology and promoting the training of users and citizens with critical thinking and principles. ethical.

· Generation of educational resources to work on AI in the school context.

In order to accelerate the transformation, we will provide resources to level children and adolescents in the basic knowledge and skills required in the development of digital competencies and AI. These resources will allow them to face industry 4.0 as users and creators, having the possibility of taking a critical look at the uses and applications of this knowledge.

Access to educational resources will be open, using various pedagogical methodologies to help students. For the above, support methodologies that have been successful at a comparative educational level will be used, so that they serve as support for the updating process of teachers and educational communities in the national context.

· Reformulation and monitoring of the school curriculum to incorporate skills necessary for the development

AI development.

The digital change is advancing rapidly, and future generations must be incorporated into this circuit from the earliest years. To this end, we will promote actions that aim to develop creative and computational thinking and awareness about electronic systems from the early stages.

learn, progressively, just as other essential skills are taught such as

those associated with language. The objective will be to promote the development of skills of

specific thinking to AI and, also, prepare students for the use of new technologies, global communications, managing large amounts of data and developing critical and creative thinking around its development and use.

· Formation of the educational community goes into skills necessary for AI development.

Although students will have to update themselves, the context will also require educators to train and acquire new knowledge and skills, while increasingly interacting with technology inside and outside the classroom.

Along these lines, we will coordinate the actors that are part of the educational community, raising awareness among institutions, and providing incentives so that they can update their skills.

Furthermore, given the speed at which technology advances and the urgency that exists in acquiring the skills around it, we will promote progress towards a more flexible system that allows experts from outside the educational communities to become part and contribute with their knowledge. , through the promotion of spaces for exchange and cooperation between regional universities and educational communities of basic and secondary education.

· Generation of participation spaces at the national level for basic and secondary education students, based on the development of projects and solving challenges.

We will create collaboration platforms with girls, boys and adolescents as the audience and main authors, since the spaces of interaction between different establishments Educational foundations create a beneficial environment to show how your ideas are being shaped, which contributes to peer-to-peer feedback.

OBJECTIVE 1.1.2

Evaluate, incorporate and promote AI as a transversal discipline in professional and technical-professional training.

· Develop AI program in CFT and IP to raise awareness and then train in AI.

According to the National Productivity Commission, in 2017 the Technical Professional Higher Education Establishments (ESTP) hosted 44% of the system's undergraduate enrollment (Arroyo and Pache-co, 2017). Furthermore, according to a recent study, the technical careers with the greatest employability were those related to technology and administration (Undersecretariat of Higher Education, 2020). It is expected that emerging careers in the area of AI will be those related to machine training, such as in the health area, or to the safeguarding and protection of data, such as cybersecurity, among others.

CFTs and IPs have already noted their motivation to integrate this knowledge to get ahead in industry 4.0 (Tomarelli, 2020). These institutions are essential to begin the updating and reconversion of workers, mainly in jobs related to automation. An example of this is the participation of State CFTs and private CFTs and IPs in the pilot of the Professional Technical Qualifications Framework in the Maintenance sector.

4.0 training in accordance with the Law on Higher Education No. 21,091. The ESTP, thanks to its configuration, has the flexibility to adapt more efficiently to these new challenges, thus training the future workforce.

Furthermore, the growing adoption of technology in the industry will require that people, regardless of whether their career is in a technological area or not, have basic notions about technology and human-machine interaction. The above becomes necessary, and indispensable, to raise awareness about AI, and then train in the same area.

For this reason, we will integrate a public-private table to incorporate AI into the CFT and IP curricula, which will also promote initiatives related to vocational and labor guidance, which will make visible the opportunities they present.

the area of AI for students, and motivate them to follow training-labor trajectories associated with the development and use of AI systems.

We will encourage the board to initially evaluate the careers in which AI is incorporated as a transversal discipline, as well as the initiatives that allow ESTP teachers to train in

digital abilities and, in particular, in AI. Along these lines, programs such as training scholarships for managers and teachers promoted by the Ministry of Education and CORFO will be promoted as part of the IP-CFT 2030 initiative, which incorporates tools and knowledge of innovation and technological transfer (Ministry of Education , 2020).

· Identify, design, adapt and enhance certification or qualification instruments for skills related to AI.

The demand for skills related to data management, high-performance computing, management and deployment of cloud platforms and AI has been steadily increasing. However, for employers requiring these types of skills, classic diplomas and certifications are not enough. This is

adds the complexity of an area that is extremely changing and advances at great speed, making it difficult to agree on the content in the evaluations.

This scenario poses the challenge of identifying the necessary competencies, designing and organizing the certification of this type of learning, or designing dynamic platforms that allow qualifying experience and skills, just as today many entities carry out ad-hoc tests to their hiring processes.

We will design new ways to certify knowledge that goes beyond traditional diplomas—such as professional certifications, university or professional technical degrees—and that changes the classic certification that indicates that attendance at a certain curriculum has been completed, for a new one that, instead, certifies experience, skills and competencies.

In addition, this design will include the formation of training and work trajectories in skills related to technologies and AI, for the harmonious and efficient development of the training process of people throughout their lives, recognizing the knowledge previously acquired.

OBJECTIVE 1.1.3.

Promote the training of skills, knowledge and aptitudes for the use, development, understanding and critical analysis of AI to all Chilean workers.

· Strengthen training programs and instruments to update and reconvert *(upskilling* and *reskilling)* to all Chilean workers.

One of the great challenges in the field of talent will be to train the current workforce, in order to avoid the obsolescence of knowledge and provide the necessary skills to respond to the digital transformation, especially the automation of tasks. Therefore, we will create training programs and instruments focused on AI, which will be the main engine to drive digital transformation in companies, increasing productivity, and improving improving the quality of life of workers. Along these lines, we will design, update and focus programs and instruments that, among other things, are responsible for: training the different groups that make up society in the face of automation; promote continuous training in companies; and provide open and free tools for the general population. It is worth mentioning that these actions will promote existing programs such as the Digital Talent initiative, building on what has been done and taking advantage of learning, networks and

the prestige achieved.

OBJECTIVE 1.1.4.

Promote the creation of specialized AI programs in the curricula of HEIs and, at the same time, their incorporation transversally across the different disciplines.

· Incorporate AI as transversal knowledge into HEI careers.

AI is required across different careers given its ubiquity across disciplines and industries. Considering the above, at the undergraduate level, without prejudice to the need to strengthen technical training in use and development in AI, training in different areas must also be adapted, or exist as complementary training.

We will establish incentives for HEIs to create multi-departmental initiatives, continuous training programs, platforms that provide challenges and tools to students, among other things. For example, professionals are needed in legal aspects of AI, in continuing or specialty training, in the social sciences. cials and humanities, in areas of business, innovation and entrepreneurship, among others.

· Training and strengthening of careers linked to computer science, robotics and associated hardware.

ced to AI.

To promote research, development, innovation and entrepreneurship around AI, it is necessary to strengthen careers linked to computer science, robotics and hardware associated with artificial intelligence.
Along these lines, we will generate incentives for HEIs to increase and improve the offer of courses associated with AI, including new curricular frameworks, and taking into consideration the creation of professional trainers, explainers and sustainers (Wilson et al., 2017).

OBJECTIVE 1.1.5.

Increase the number of AI experts, that is, masters and doctors, to a value equal to or greater than the OECD average and create incentives for the incorporation of this talent in both the academic, State and private sectors.

· Focusing public funds aimed at training talents linked to research and technical work around AI.

In order to strengthen the institutions research, technology centers and companies, it is necessary to have coordinated action between the training of talents, and an adequate insertion of this new generation of researchers. These talents are indispensable and responsible for being a driver of innovation and change within organizations.

The evidence shows how the insertion of researchers into the industry has a positive effect on productivity. This is how there are already efforts prepared for this goal such as the National Policy of Science, Technology, Knowledge and Innovation, led by MinCien-cia, which includes the training and insertion of talents in the productive structure and in the country's academy, also taking into consideration the strengthening of local master's and doctoral programs.

Along these lines, we will implement targeting actions and design mechanisms for studies in areas related to AI in China.

him or abroad, according to the pilot implemented in 2020 that focused Chile Scholarships for doctorates abroad. At the same time,

To take advantage of the potential of Chilean HEIs and their prominent position in the region, instruments will be created to attract talents and thus strengthen national programs.

Finally, we will work together with the industry to promote investment in talents in the private sector through agreements and design of new shared financing instruments.

· Generation of incentives in the

demia and the local industry to take advantage of talents trained in AI.

In view of the growing insertion of new technologies to the national ecosystem, it is essential to retain talent and investment in the country's education. Therefore, the focus should not only be on increasing the number of people trained in AI, but also on generating a stable environment over time capable of offering competitive opportunities for academic and professional development.

Along these lines, and in agreement with the private sector, we will make the national ecosystem more attractive and encourage the insertion of experts in industry and academia.

Additionally, we will take advantage of those who decided to pursue their career abroad, generating networks of national professionals and experts that contribute to the development of the local ecosystem of science, technology, knowledge and innovation.

· Adapt the accreditation of the national programs according to international AI metrics.

AI and the disciplines associated with its development respond to incentives and have ways of measuring their success and development that are different from other areas of knowledge. Analy- We will review the international state of the art, to design appropriate indicators for accredit programs associated with computer science, electrical engineering and AI in general.

Thus, we will contribute to the review and update of the accreditation of HEI programs by the CNA, while we will establish incentives for the implementation of initiatives.

OBJECTIVE 1.1.6.

Coordinate the national system of science, technology, knowledge and innovation so that the trained talent points to the future we long for.

· Permanent demand prospecting exercises.

We will maintain a registry that quantifies and projects the needs for talent specialized in AI and other technologies, clearly identifying the demand for the public, private and academic sectors.

· Permanent anticipation exercises.

We will coordinate anticipation exercises on a permanent basis with a view to promoting the creation of desirable scenarios for the country and containing risky scenarios through the training and insertion of talent.

· Constant updating of talent training roadmaps.

Based on the baseline presented by the demand prospection and the technological scenarios, opportunities and risks identified in the anticipation exercises, we will coordinate the creation of roadmaps regarding talent training, which specify needs, raise and direct financing instruments, necessary collaborations, and call for the execution of public policies, measures and required investments

1.2. TECHNOLOGICAL INFRASTRUCTURE

Technological infrastructure is a necessity to improve the productivity and development of the country, integrating in a respectful, safe and quality manner into the reality of each territory. The development and use of AI requires adequate connectivity, ubiquitously accessible platforms (such as the cloud), data storage centers, among others.

The technological infrastructure not only impacts through the improvement in productive processes and work methodologies, but also in the final service that the population receives and the facilities that they mean for their daily lives. An example is the implementation of 5G, which has meant large investments in recent years to support digital transformation.

Currently, Latin America is far behind compared to the northern hemisphere and Oceania in terms of required infrastructure, but in a growth phase that involves the arrival of new technologies that are contributing to the well-being of the population, and that have been fundamental , for example, to maintain the continuity of activities during the pandemic. These activities have created a growing

infrastructure demand.

In this sense, Chile stands out in Latin America and the Caribbean for internet access, more than 82% of the population has access to the internet, which makes it one of the countries with the greatest coverage with a Digital Adoption Index. of 75.62 according to the World Bank (Gómez et al., 2020). In addition, there is a growing presence of data centers from large technology companies, processing infrastructure such as the National Laboratory for High Performance Computing (NLHPC) and public-private initiatives such as the Data Observatory, which takes advantage of the unique conditions in a scientific field such as astronomy to promote the training of talent and the use and development of systems for the analysis of large volumes of data.

Despite the above, Chile still has to resolve important gaps, accelerate the deployment of infrastructure and capitalize on existing opportunities to obtain greater benefits from AI. For example, according to a study carried out by the Undersecretary of Telecommunications of the Ministry of Transportation and Telecommunications (Subtel), only 57% of homes have residential internet, and due to the high concentration of the country's population In urban areas, a significant percentage of the national territory does not have sufficient connectivity for the use and development of AI.

Chile does, however, have the value of multiple Natural Laboratories, that is, sites with unique geographical characteristics that enable competitive advantages for the development of science and technology at a global level (Guridi et al., 2020). This is of great relevance because it shows that there are areas of the country that are worth the effort to connect to study them and put the knowledge of global value that is generated there at the disposal of humanity.

Finally, as mentioned in the previous section, in parallel with the development of AI, other technologies are evolving at high speed and with the potential to generate a relevant impact on their nature and deployment.
In this sense, we must maintain anticipation exercises so that this Policy continues to be aligned with the future of AI and other technologies yet to be known, also in relation to infrastructure.

To address these challenges, the following objectives are proposed:

OBJECTIVE 1.2.1.

Turn Chile into a global hub for the southern hemisphere in technological infrastructure that is at the global forefront in AI.

· Acceleration of the deployment of the National Connectivity System.

Although Chile has a good infrastructure in terms of connectivity compared to countries in the region, challenges persist in areas far from the national capital and in the provision of the last mile.

Consequently, we will facilitate, on the one hand, the accelerated progress of fiber optic tenders in the northern and southern zones and their timely construction to provide backbone connectivity and, on the other, the implementation of last mile and 5G solutions that enable the access to the highest bandwidths[2] .

· Formation of alliances in America Latin America and the Caribbean for the implementation of projects of regional and global importance.

We will promote initiatives for the development of infrastructure of regional importance such as the digital bridge between Asia and Latin America and the Caribbean. Given the leadership position that Chile has in this area in the region (Gómez et al., 2020), it seeks to consolidate its comparative advantages and thus attract investment and facilitate the internationalization of local developments.

In addition, we will open the opportunity to incorporate compensation at the regional level for the development of science and technology both in Chile and in alliance with other actors from Latin America and the Caribbean in the bidding processes.

2 5G is the fifth generation of mobile networks that enables a leap in speed, capacity and response times in communication between mobile devices.

This will have important consequences on different aspects of daily life, security and the economy around the world. 5G is expected to promote innovation in markets, improve the performance of the economy and the quality of life of citizens.

OBJECTIVE 1.2.2.

Deploy connectivity infrastructure that guarantees access with high quality standards for Chileans.

· Generation of citizen connectivity projects and state subsidies.

To promote the use of AI and other technologies, we will work on agreements and standards to achieve minimum quality in access to connectivity for citizens.

In this sense, we will promote public Wi-Fi type projects, while at the same time providers will be required to reach consensual quality minimums. In addition, we will establish the connectivity of universities and R&D centers as a priority when bidding and implementing last mile projects, in conjunction with the national university network (REUNA).

To attract investment in less populated areas, the State has subsidies specially designed to cover these gaps. Along these lines, we will identify areas that require subsidies and connectivity projects to guarantee a minimum of access to all citizens.

· Modernize regulation to guarantee minimum standards of access and quality of service.

We will promote an analysis of the institutions and current regulations so that the necessary powers to supervise suppliers exist in the State, which are diffuse or non-existent in the

current institutions.

In addition, we will contribute to establishing compliance standards in service quality based on international examples agreed upon with suppliers. In this sense, we will promote the empowerment of Chileans as inspectors of the quality of the internet services they receive.

OBJECTIVE 1.2.3.

Deploy technological infrastructure that increases storage and processing capacities in the country.

· Promotion of private, public and public-private investment in technological infrastructure based on Chile's comparative advantages.

We will promote initiatives that take advantage of our comparative advantages at a global level to attract public and private investment

including our Natural Laboratories, and we will promote collaboration in the installation of scientific-technological infrastructure in international

collaborations; Examples of the above are the Astronomical Park or scientific development.

Chajnantor

co, technological and Antarctic logistics.

OBJECTIVE 1.2.4.

Coordinate the science, technology, knowledge and innovation ecosystem to understand the technological infrastructure needs associated with AI that the future we long for requires.

· Permanent exercises of prospecting and anticipation of infrastructure demand for country challenges.

We will maintain a registry that quantifies and projects the need and use of storage and computing infrastructure, high-performance computing or related infrastructure by the national ecosystem of science, technology, knowledge and innovation.

We will coordinate anticipation exercises on a permanent basis with a view to promoting the creation of desirable scenarios for the country and containing risky scenarios through investment in technological infrastructure.

ca, establishing innovation challenges and roadmaps that specify needs, raise and direct financing, necessary collaborations, and execution of public policies, measures and investments.

The roadmaps will design and implement mechanisms to attract private investment in technological infrastructure, recognizing that the main capabilities are found in companies in the sector.

To this end, we will promote the demand for technological infrastructure services, the evaluation of subsidies for access in areas of lower demand, and the generation of multi-sector alliances.

1.3. DATA

The COVID-19 pandemic has contributed to assessing scientific evidence in the decision-making of decisions and the availability of good quality open data, correctly organized, in the appropriate volumes and formats. The latter has been important for the knowledge and control of the virus and its impact on public health and the economy, allowing timely and robust decisions to be made that have facilitated the work to overcome the emergency. For AI, data constitutes an essential requirement (COVID-19 Data Sub-table, 2020).

For there to be an effective deployment of AI in Chile, an ecosystem is necessary.
ma where there are open and high-quality repositories, but that are also safe and protect people's rights, this is why it is relevant to promote models that strengthen trust and the conditions to encourage multiple organizations and/or companies to share data in favor of the common benefit.

Although the existence of large volumes of data is not a guarantee of good AI algorithms, there is consensus regarding the convenience of having them for adequate training of the algorithms, even reaching the point that a mid-level algorithm reach or surpass a first-level one if it has a greater volume of data (Lee, 2018).

However, not only volume is enough, but it is required that the data adequately represent reality (be of good quality), with levels of controlled bias, interoperable, correctly anonymized and aggregated (if applicable), among others. characteristics.

Having Natural Laboratories allows our country to build a path of development of data repositories unique in the world, thereby attracting advanced capabilities for AI to Chile, which include talent, institutions and top-level investments in the global scenario.

Finally, although the use of data does not necessarily target those that can be associated with an owner, it is necessary to update the regulation on the use of personal data with a view to reducing uncertainty and achieving international standards. In

Along these lines, we must have a reformed personal data protection law and also cross-cutting institutional and technical developments to allow the advancement of technology, while at the same time the rights to privacy, the protection of personal data and non-arbitrary discrimination be cautious.

To address these challenges, the following objectives are proposed:

OBJECTIVE 1.3.1.

Promote and consolidate a data agenda of public interest, which results in both legal certainties and clear definitions of responsibilities within the State, and that promotes a public-private ecosystem of generation and access to quality data for the use and development of AI and related technologies.

· Update the regulation on personal data and generate mechanisms that allow adaptation to new technological developments.

We will promote pending legislative work regarding Personal Data, incorporating what we have learned in Europe with the General Data Protection Regulation (GDPR).

We will generate pilots that allow technical evaluations and enable spaces for dialogue and exchange to promote differentiated analyzes in the State and the expertise private sector, and we will promote alliances between the private sector and the public sector with industry actors who have data of value for the development of science, technology, innovation and entrepreneurship, and services such as MinCiencia, the Ministry of Economy and/or the Ministry of Labor.

We will promote research and the use of systems that allow data to be processed in a distributed and secure manner.

Also, we will promote the research and development of anonymization, bias identification and computational security techniques that allow the use of data in accordance with the principles that guide this policy.

· Create and consolidate adequate data governance in the State that promotes data availability

quality.

The State manages a large amount of administrative and population data, which can be used by government agencies, academics and private organizations, to generate knowledge and solutions that contribute to the quality of life of citizens.

For this to happen, the State must be able to make available, complying with the regulations and the protection of personal data, the greatest amount of data possible, ensuring that it is done in a timely manner (that is, when it is required), high frequency and periodicity, easy to interpret, reliable, with an adequate standard to be worked on and interoperable, so that they can be used from different services.

The data made available by the public sector show important gaps in meeting these characteristics. Although there is a centralized data platform for the public sector, called Open Data Portal (datos.gob.cl), of the Ministry of the General Secretariat of the Presidency, institutions use it only partially and sporadically.

On the other hand, given the sensitive nature of some data managed by the State, its publication must be done with standards of informed consent and there is an obligation to protect personal and sensitive data. This does not imply storing them in a closed manner and generating inaccessible repositories, but rather generating administrative and technological mechanisms through which this data can be worked with without violating people and their privacy. Along these lines, we will design adequate data governance in the State

that promotes the availability of quality data, on the appropriate platforms, in the best possible order, with interoperability standards, in a secure manner, properly aggregated and/or anonymized if applicable.

· Promote the development of institutions

tutionality, trust and mechanisms for the industry to share and make available data at the sector level.

Aggregated and anonymized data at the level of industrial sectors and/or sufficiently large territories can generate high-value repositories that boost the productivity of all companies in the sector, in addition to opening spaces for research, development, innovation and enterprise. arrest. This poses challenges at a technological, institutional and trust level, since mechanisms must be agreed upon to share data, safeguarding the rights of people and the competitiveness of companies. In this sense, we will analyze the current institution and work with public and private industry to advance

The matter. This includes designing appropriate incentives, building trust, promoting through ANID and CORFO instruments as appropriate the development of common repositories at the economic sector level.

mico or industry and even initially launch these platforms.

Thus, we will promote a data ecosystem of global value from Chile, with sources of high merit for the development of cutting-edge knowledge, public, open and secure data for the development of AI, which encourages the development of talent and technology. and attract efforts from other global AI development partners in the northern hemisphere and Oceania. hubs

· Launch initiatives that act as catalysts for the ecosystem.

topic in priority areas, based on the country's strategic advantages.

We will launch entities, such as the Data Observatory and the Climate Change Observatory, which will be catalysts for the creation of talent and technological solutions for access to large volumes of data.

These entities will have as their central mission to promote the ecosystem and its capabilities, using as their main asset the appropriate availability of data sets that Chile has and are unique on the planet, such as those of our skies that allow advance global astronomy in the north of the country, or those of our oceans and cryosphere that allow progress in the adaptation and mitigation of Climate Change.

· Promote the availability of scientific data.

The availability of scientific data is relevant for the advancement of science, and of special importance for the replicability of the experimental results obtained, in accordance with the FAIR3 principles.

We will advance in the implementation of technical and legal mechanisms that allow greater availability of scientific data to promote collaboration, in addition to allowing these data to be used by citizens and the productive sector.

The above is not without challenges, since it requires the coordination of various actors, the protection of legitimate interests, legitimate—such as embargo periods so that the researchers who extracted the data can publish—among other factors that have been identified by ANID in its Open Data Policy (ANID, 2020).

· Promote the creation and activities of committees or user communities of data of public interest.

We will promote the growth of communities of users of public interest data, with a focus on the creation of trust, standards and agreements between data producers and consumers, and on the development of an ecosystem that adequately considers the various cases of use that evidence has.

3 The FAIR principles consist of a set of qualities to make data Findable, Accessible, Interoperable and Reusable.

AXIS 2.

DEVELOPMENT AND ADOPTION

In Latin America and, in particular, in Chile, there are important challenges related to research, development, innovation, entrepreneurship and adoption of AI.

To increase our country's productivity and position ourselves as an AI leader in the region, our AI economy must be much stronger. Along these lines, in this axis we will address the factors that allow the increase in the supply and demand of AI, understanding it as the development of solutions based on AI, and the adoption of these to increase productivity in the generation of knowledge, cultural elements, products or services.

Currently, the global impact of R&D done in Latin America and the Caribbean is minimal. In fact, the AI Index Report prepared by Stanford University (2019) reports that this region contributes less than 5% of publications in academic journals and conferences (see figures 3 and 4), and generates less than 2 % of citations for both academic journals and conferences. Regarding patents related to AI, the panorama is more critical, less than 1% correspond to Latin America and the Caribbean (Stanford University, 2019). Consequently, private sector investment levels in AI in the region are much lower than those recorded in the United States and China (see figure 5).

FIGURE 3: Percentage of AI publications in academic journals by region between 1990 and 2018. Source: AI Index Report 2019.

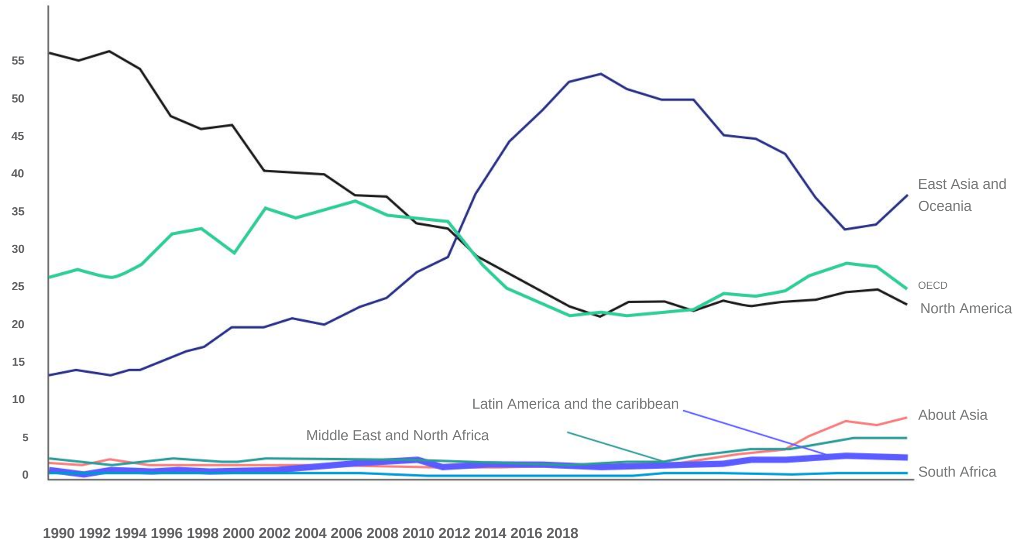


FIGURE 4: Percentage of AI publications at conferences by region between 1990 and 2018. Source: AI Index Report 2019.

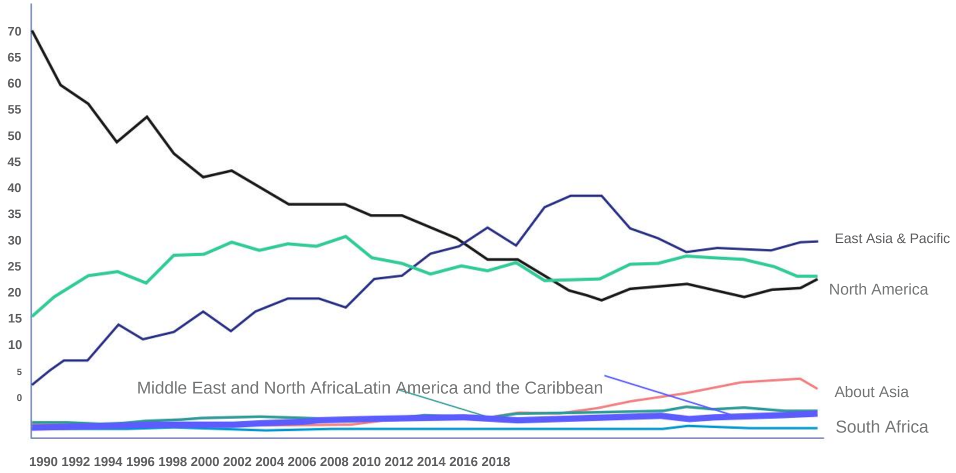
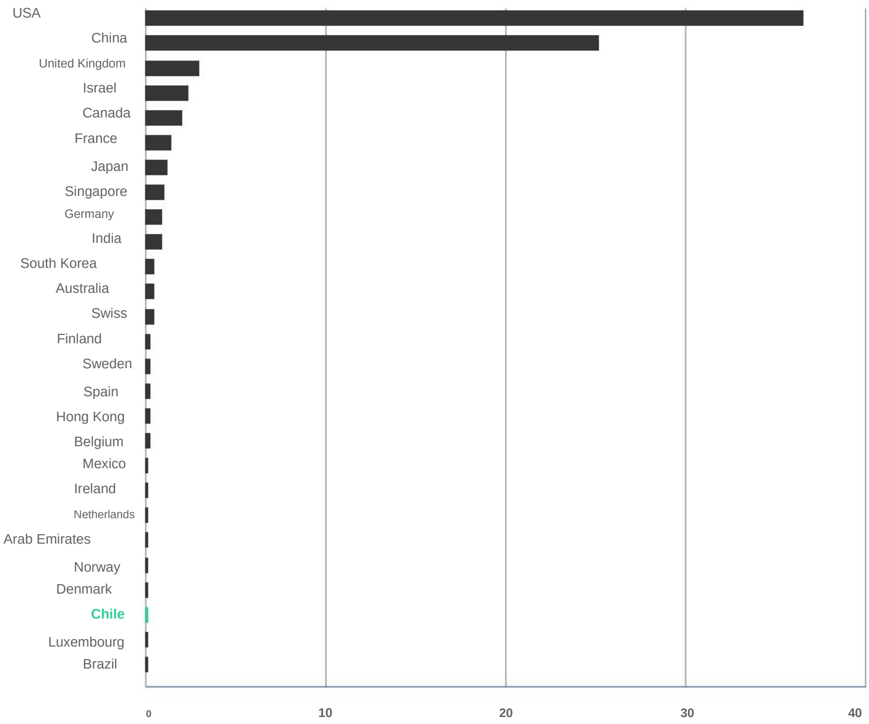


FIGURE 5: Total investment in AI from January 2018 to October 2019 in trillions of current dollars. Source: AI Index Report 2019.



In our country, the adoption of AI and its impact on productivity is still low and far from the levels we need to achieve the objective of this policy. An exploratory study on companies carried out in Chile by the Chilean North American Chamber of Commerce (AMCHAM) and the Universidad del Desarrollo, finds that 78% of the companies surveyed do not use the data they have to make decisions and are very far from im- implement prescriptive intelligence systems based on them (Chilean North American Chamber of Commerce and Development University, 2019).

Consistent with the above, in the latest Information Technology survey of the Ministry of Economy, we found that only 2.2% of companies use data analysis (Ministry of Economy, Development and Tourism, 2020).

Finally, Chile does not currently have indicators and baselines on the use and productivity of AI. We need to create them as they constitute an objective resource to know the progress in the proposed initiatives and compare ourselves with other countries.

To address these challenges, the following objectives are proposed:

OBJECTIVE 2.1.

Generate AI productivity indicators for Chile.

· Generate indicators of scientific productivity and technological transfer of AI.

There are currently no adequate indicators that monitor scientific production and technological transfer related to AI. Along these lines, we will build the necessary indicators in accordance with international experience, to then maintain continuous monitoring and updating. To contain these indicators we will use "Observa", the Science, Technology, Knowledge and Innovation Observatory, which will keep this information available.

· Generate AI productivity indicators in the economy.

Currently, there is no official, reliable measurement of AI productivity in our economy.

Consequently, we will build them based on international experience and use existing instruments such as the this TIC survey of the Ministry of Economy, Development and Tourism, applied by INE for the first time in 2018 to collect the required information. We will encourage the adoption of these indicators by INE.

· Generate AI productivity indicators in the State.

Currently there is only one indicator that allows measuring the adoption of AI in the State, which corresponds to the survey applied by the Digital Government Division with respect to multiple technologies. Along these lines, we will promote and improve this instrument so that it measures productivity, and collect information for indicators in a more robust and periodic manner. Furthermore, given the existence of initiatives such as the

OECD AI Policy Observatory, we will design a national census that makes visible the use of AI in the different re-partitions of the State, which is constantly updated.

OBJECTIVE 2.2.

Strengthen Chilean R&D in AI to achieve a level equal to or higher than the OECD average.

· Modify accreditation criteria

for AI-related programs that adapt to each discipline.

Currently, the criteria used to accredit and evaluate the performance of researchers related to AI do not adjust to their work, nor to the dynamics of the academic community involved.

international. In line with what was proposed in the previous section on “Talent Development”, we will propose modifications to the way in which the performance of researchers in AI, which is under the mandate of the CNA, is evaluated, so that it adjusts to the current scenario. Likewise, we will maintain monitoring of the internal reality

international and a constant and periodic update of the accreditation criteria, in support of the work carried out by the CNA.

· Promote R&D in AI, both in academia as well as in industry.

Advances in AI occur in both academia and industry, and involve multiple disciplines in addition to engineering and computer science.

We will allocate or focus resources towards

It promotes the disciplines associated with AI and/or that generate applications using its systems, or towards attracting more researchers to both industry and HEIs. Along these lines, we will promote public-private agreements that allow promoting R&D in AI in the productive sector.

OBJECTIVE 2.3.

Promote collaboration between academia and the productive sector for R&D of systems with AI.

· Promote the development of joint industry-university projects.

Some of the most important applications of AI have been developed in HEIs, giving rise to successful products and companies.

In accordance with the above, we will prepare a registry and make visible the existing success cases of collaboration projects between the productive sector and academia. Likewise, we will design instruments and institutions that strengthen and foster long-term relationships between the productive sector and the academy, which will allow the generation of projects with a shared vision with benefits for both parties.

· Promote and enhance postgraduate programs that incorporate objectives of the productive sector.

In line with the previous action, an effective knowledge transfer mechanism

ment are joint postgraduate programs between universities and companies.
In Chile, AI programs are beginning to emerge in different institutions.
These programs must be consolidated as

real spaces for collaboration, so we will promote their accreditation and design incentives that stimulate the participation and active involvement of students in companies.

· Promote the insertion of talents in the productive sector.

In the field of AI, there is a significant demand for experts at an international level and, if once trained in Chile there are no adequate conditions for their insertion, it will be difficult for them to contribute locally, running the risk of talent drain.

Along these lines, we will promote innovation in the productive sector and review competitive insertion mechanisms at an international level so that professionals associated with the fields of AI contribute to the local industry. In addition, we will promote greater flexibility in the academy so that university professors can contribute to the productive sector without seeing their academic careers harmed and with adequate recognition for the productive development that their contributions generate.

OBJECTIVE 2.4.

Promote the development of the R&D&i ecosystem in AI where the State, the productive sector and academia generate, are attentive and invest in opportunities related to AI.

· Anticipate challenges and missions that guide the work of the ecosystem.

AI opens up the opportunity to be pioneers in emerging markets through prospecting, anticipation and prioritization exercises. We will raise the use and development of multi-sector AI to prospect its evolution and we will coordinate anticipation exercises.

The results of these actions will serve to identify, from the different territories of the country, missions and challenges that will guide the development of the ecosystem strategically, taking advantage of the comparative advantages of each territory.

· Promote the development of scientific-technological-based ventures

logic with AI.

Scientific-technological-based enterprises (EBCT) are different from other types of companies, since they are created on the basis of knowledge with innovative potential, which arises from the R&D carried out within the company. academia or industry

(Ematris, 2020). It is for this reason that to promote them, specialized instruments must be available that adapt to their reality and aim to generate minimum conditions for their existence and survival.

We will push the investment in AI EBCT to a level equal to or higher than the OECD average, firstly, observing the results of the previous priority action and complementing it with records of EBCT success cases that may emerge to study their needs.

Next, we will review the State's development instruments and promote public-private agreements, including attracting foreign investment so that they respond to those needs.

In addition, we will promote the existence of risk capital for EBCTs in their different stages, from seed funds for prototyping to expansion and internationalization. The case of Israel, which is approaching investment from the European Union (OECD, 2019), is an example to follow.

· Generate a community of

entrepreneurs and innovators in AI.

To encourage investment and promote entrepreneurship based on AI, a community must be consolidated and strengthened around R&D and use of technology. Along these lines, we will generate channels and events that allow us to build networks, share experiences, make success stories visible and attract investors. This is combined with strategies to promote the adoption of technology in companies, since through these communities and the strengthening of the ecosystem, barriers will be reduced, accelerating the

digital transformation.

OBJECTIVE 2.5.

Promote and boost the economic productivity of AI to reach levels equal to or higher than the average economic growth for OECD countries due to the impact of AI.

· Training for boards of directors, executive levels and union leaders of the productive sector.

A deficit detected in the productive sector is the low involvement and high ignorance of its leadership about the potential and nature of the use and development of AI. The above is also seen in unions and associations. This generates many difficulties in the required transformations; it also happens that many organizations feel the need to implement AI systems without requiring it for their objectives or, worse, without adequate management of the associated risks.

Consequently, we will create and promote training opportunities for leadership roles in the productive sector, appropriate to their desires and sensitivities.

· Generation of incentives and promotions for the adoption of AI in the industry.

Based on the data collected by the survey ICT, the challenges and the defined missions, we will review the incentives and training instruments

ment for the use and development of AI in the country, and we will promote the creation of new ones to strengthen the ecosystem. This will contribute to promoting the adoption of technology in a scenario of economic recovery after the COVID-19 pandemic.

· Make the use of AI visible in the industry.

The dissemination of success stories and greater dissemination of the topic is vital for decision makers in companies to accelerate their implementation and trigger an increase in investment in natural solutions.

tionals. To do this, we will implement a platform that brings together initiatives to make local developments visible, which will also have a coordinated dissemination plan between the communications units of all participating ministries.

OBJECTIVE 2.6.

Accelerate the modernization of the State through AI.

· Missions and challenges of innovation in the State.

Based on the data collected by the Digital Government Division survey, the progress in the implementation of AI systems in different departments, the projects detected by the proposed census, and the results of the anticipation exercises, we will guide and we will promote the digital transformation processes required by the public sector for the implementation of AI with an impact on its efficiency and productivity.

· Training and incentives to promote AI in the State.

The implementation of AI systems in the public sector depends, to a large extent, on properly aligning the incentives of public officials with citizens. Along these lines, we will design management instruments that generate adequate incentives in people for the implementation of AI systems in the State, such as Management Improvement Programs (PMG).

Just as there is ignorance among private sector managers, this is also a reality in the State. The problems are similar, since systems are purchased without knowledge, which are not applied adequately, opportunities are lost, there is no notion of the transversality of the technology, among others. However, the risk in these applications is greater, since poorly used technologies ceded in the State, or lost opportunities, mean potential loss of benefits or harm to society. Along these lines, we will articulate AI training for public officials, both at the level of authorities, managers, and technical teams.

· Modernization of public procurement processes for AI systems.

The implementation and adoption of AI systems in the public sector depends, to a large extent, on having an adequate system to acquire them, which is observed in the Public Procurement regulations. Along these lines, we will modernize Public Procurement processes, so that they allow AI systems to be acquired and implemented correctly.

At this point it is relevant to highlight the generation of incentives for the purchase of nationally developed systems.

Thus, public purchases allow the local ecosystem of innovation and entrepreneurship to be encouraged, and the State can opt for solutions adjusted to the national reality.

· Valuation of success stories in the adoption and use of AI in the State.

Furthermore, there is no coordination between different public services for the implementation of AI systems and transfer of best practices. Consequently, we will make visible the success stories that emerge.

These examples will serve as learning when designing and implementing new projects.

OBJECTIVE 2.7.

Adoption in key challenges: mitigating and adapting to Climate Change with AI.

· Promote research, development and use of environmentally friendly AI systems.

The computing power used in training algorithms has doubled on monthly time scales in recent years, so it is estimated that by 2030 half of the world's electricity consumption will be due to this activity (National Institute for Research in Digital Science and Technology, 2020).

Consequently, we will establish incentives so that the development and adoption of AI in Chile focuses on energy efficiency. The above will be possible by carrying out a review of Law 19,300 of General Bases on the Environment, regarding energy efficiency of technological systems based on AI.

In addition, we will incorporate environmental considerations into assessment instruments.
promotion and good practice guides will be developed —based on pilots—for environmentally responsible use in the public and private sectors.

· Promote the use of AI tools for timely, effective and efficient monitoring of the environment that contribute to reducing the environmental impact of the State and the industry.

We will promote standards, interoperability and access to critical data about our environment, aiming to make monitoring compliance with environmental regulations more efficient and productive, but also to begin generating predictive analyzes that allow catastrophes to be prevented.

For this, we will articulate standards and gather data from multiple sources, capturing it in real-time, in a massive and automated way. This will allow, for example, the Environment Superintendence (SMA) to activate timely alerts and activate actions by the competent State bodies.

Along the same lines, we will promote that citizens have adequate access to relevant and real-time information on climate change in their territory, through a Climate Change Observatory (OCC).

AXIS 3.

ETHIC,
LEGAL ASPECTS
AND REGULATORY,
AND IMPACTS
SOCIOECONOMIC

3.1. ETHIC

Although AI offers opportunities and benefits to society, there are uses of this technology that present risks associated with fundamental rights such as dignity, privacy, freedom of expression and non-arbitrary discrimination.

That is why it is essential that its development considers reflections regarding the cultural and ethical problems that it can generate, and risk mitigation actions to implement AI in a responsible manner. Due to its rapid evolution, and its increasing incorporation into daily life, these analyzes go beyond legality, and motivate a social debate: which uses are beneficial and which are not for individuals, communities and/or society in general; Human-computer interaction raises opportunities for collaboration, but at the same time new questions and impacts.

Some of the issues raised in this area are the presence or amplification of unwanted biases in algorithms and the databases that feed AI systems; the development of surveillance systems based on recognition

the need for consensus in applications such as autonomous vehicles (Awad, 2018), from moral dilemmas in fatal accidents to cyber attacks on these technologies (IDB, 2019); the impact of automation in the workplace (Author, 2018), in which there will be risks associated with loss of jobs in tasks that can be performed by a machine, but at the same time many new ones will be generated roles (WEF, 2020), among many

Other themes.

This scenario makes it especially important to consider the concerns, advantages and opportunities that people perceive about this technology and the solutions based on it (Hidalgo, 2021). In this sense, the search for solutions through global standards and principles faces the multiplicity and diversity of cultures and idiosyncrasies in which AI systems are immersed. For this reason, the answers to ethical crossroads are not unique, given that the technical is coupled with a social and political debate that must be promoted at the local level. Therefore, local cultures and idiosyncrasies must be taken into account.

biometric, which when misused could put our personality in check in the digital world (Garrido and Becker,

2017); the appropriate balance between privacy and explainability with efficiency and effectiveness (Kearns, 2019), so as not to compromise the right to privacy to the detriment of certain groups in society;

taken into account when developing national AI policies and strategies.

However, to date, the discussion that has led to the creation of a catalog of principles that should govern AI has been concentrated in the northern hemisphere, particularly in the United States and

Europe. This is evidenced in the Harvard Berkmann Klein Center publication: of 36 AI principles identified in the world, only two have their origin in Latin America (Fjeld et al., 2020). Another publication analyzes 84 documents with ethical guidelines, none of which come from Latin America or Africa (Jobin et al., 2019). The draft Recommendation on Ethics

of UNESCO AI submitted for public consultation until July 31, 2020 expresses this concern:

"Emphasizing specific must be provided in that attention to the lower middle, the countries of entry included, the among others, Africa, America Latina and the Caribbean and Asia, Central, as well as islands in small developing states, have already been underrepresented in the debate on the ethics of AI, raising concerns that local knowledge, pluralism the in "ethical culture, value systems and demands for global equity."

(United Nations Educational, Scientific and Cultural Organization, 2020).

This concentration of the discussion makes it extremely relevant that developing countries become part of and contribute from their local reality to the discussion of the principles that should govern AI.

It is increasingly necessary for each country to analyze whether its current standards are sufficient, whether they require modifications, or whether the promotion and generation of new standards is pertinent. In the case of our country, in this analysis we must consider the bills that allow a safe and reliable use of AI, especially those that modify laws No. 19,628, on the protection of private life. (1999), and No. 19,223, which typifies criminal figures related to information technology (1993).

OBJECTIVE 3.1.1.

Promote the construction of regulatory certainties on AI systems that allow their development, respecting fundamental rights in accordance with the Constitution and the laws.

· Develop a survey on ethical and regulatory aspects of AI.

The discussion on regulation and governance of AI is recent and is far from reaching a point of completion at a global level; there are many areas that have not yet been analyzed from a regulatory perspective.

That said, Latin American countries like Chile are particularly delayed, both in the discussion and in regulatory advances. This delay generates uncertainty when it comes to researching, developing, innovating, undertaking and adopting.

Along these lines, we will design and implement a review agenda of the existing regulations related to the fundamental rights that are or may be be impacted by AI, and we will determine which standards should be repealed or updated, in accordance with the advancement of AI.

It is important to point out that the development and evolution of emerging technologies such as AI impose the challenge of a permanent and agile review and update, so that it is possible to respond to the speed of their advances. Therefore, in this agenda we must promote an open and constant dialogue, which allows us to reach consensus and generate norms in a dynamic way that adapt to our national reality, not only adopting what has been developed in other countries.

· Become an active part of the international discussion on principles and standards, making the national reality visible and taking a leadership role at the regional level.

Emerging countries, and particularly those in Latin America and the Caribbean, have experienced low participation in the discussion and elaboration of principles that will govern AI in the coming years.

Due to the high citizen participation in the development of this Policy, we already have a long way to go. Thus

We will now seek to participate in international spaces for discussion and exchange, ensuring that they reflect the Transversal Principles of AI, for example, those of UNESCO, the recommendations of the OECD, among others.

Furthermore, it is crucial to bring positions closer together in Latin America. These will allow the international discussion to be addressed from the region and influence decisions on principles and positions. We will play a leadership role as we are one of the countries in the region with a more developed AI ecosystem.

In fact, our country appears in second place in Latin America in the AI Government Readiness Index prepared by Oxford (2020).

· Develop the requirements to ensure in an agile manner the development and responsible use of AI.

We will design an agile institutional framework that establishes the requirements or aspects that must be met to respond to changes and the rapid development of technology, promoting the development of a flexible regulatory framework and a non-overregulated ecosystem. This institutionality must maintain communication

Active with sectoral regulatory and supervisory bodies, acting as a technical informant on matters related to or associated with AI.

In the same sense, the institutions must promote collaborations, alliances and commitments in the private sector for the development of good practices in the development and ethical use of AI. The purpose of this is to respond in a timely manner to challenges, recognizing the rigidity of regulation through legal standards, providing flexibility so as not to harm development.

On the other hand, the institutional framework will promote the formulation of good practice guides based on pilots and comparative experiences, both for the State and the private sector; will identify spaces for the implementation of incentives for the good use of AI; and will explore the possibility of generating regulatory sandboxes for certain applications, among other measures.

Likewise, it is important for the proper functioning of this institutionality to have support from the academy, which will be necessary due to its specialized knowledge on the subject.

The institutional framework will contribute to guiding the development of research around relevant topics when it comes to cautioning the use of AI, such as the transparency and explainability of algorithms, the identification and suppression of biases and arbitrary discrimination, cybersecurity, among others.

Finally, this institutional framework must consider the interests and visions of underrepresented groups when addressing these issues. An example is the special concern that should be taken in AI systems that affect decisions associated with girls, boys and adolescents.

OBJECTIVE 3.1.2.

Boost algorithmic transparency.

· Establish algorithmic transparency standards and recommendations for critical applications.

The delivery of information on how the decisional algorithms used by state administration bodies work, as well as the data involved in decision-making, including those from the learning phase, must be timely and clear in accordance with the right of access established in Law No. 19,628.

Likewise, we must ensure the identification of biases in algorithms, databases and other components of the systems.

more of AI, and for the mitigation of risks of affecting fundamental rights, especially in the case of privacy, protection of personal data and arbitrary non-discrimination used by the State administration bodies.

In this sense, based on Chile's international leadership in transparency, we will develop recommendations for the private and public sectors in relation to the identification of unwanted biases and algorithmic transparency, which can be piloted in risk areas.

3.2.
IMPACTS ON WORK

Advances in the development of emerging technologies, including AI, are changing the nature of work, and are expected to continue in the coming decades. Likewise, it is demanding that new generations update the necessary skills that allow them to work and adapt in the AI era. These types of transformations have been a source of concern due to the risk that certain tasks commonly performed by humans will be replaced by machines, which would imply the loss of work for a large number of people, especially for those who perform repetitive tasks.

Furthermore, the adoption of AI not only makes it possible to automate routine tasks or tasks that require high levels of qualification that were not possible until now, but it changes the nature of many occupations.

nes. In this sense, people must not only reconvert, but also update and adapt to the growing human-machine interaction in various areas.
In Chile, employment is associated with the social protection network, therefore this risk has a greater potential impact.

However, international evidence shows that in the long term the adoption of new technologies represents an opportunity to generate new jobs and increase the productivity of a country, something that has already happened with other industrial revolutions in the course of the history (Organization for Economic Cooperation and Development, 2019). Along these lines, we will work together with the private sector to promote measures to mitigate the impact on people's lives and contribute to ensuring that the changes improve their future well-being.

OBJECTIVE 3.2.1.

Carry out prospective analyzes to actively detect the most vulnerable occupations, anticipate the creation of new jobs through AI and support workers in the transition to new occupations, minimizing their personal and family costs.

· Prospecting and anticipating changes in the labor market to
reduction of gaps.

One of the phenomena that occurs with automation is what has been described as "employment polarization", which is characterized by an increase in the hiring of low- and high-skilled sectors, to the detriment of those sectors of intermediate qualification, the sectors of

Lower qualifications are more exposed to automation risks. Even though it depends on the study and the methodology used, and without considering the effect of the COVID-19 pandemic, which has shown to have the potential to accelerate these processes, most projections conclude that the greatest impact on the country could be observed in the next two decades. Some figures indicate that Chile is one of the OECD countries with the highest average probability of automation, ranging between 42% and 52% of occupations. Others point out that the first wave of automation in the country will occur at the beginning of the 2021-2030 decade, affecting only 1% of jobs, that towards the end a second wave would affect 13% and that at the end of the decade 2031-2040 it will have reached 28% (PwC, 2018).

On the other hand, automation could also deepen inequalities both between countries and within them between occupations, educational levels and gender. In particular, in Chile the probability of automation is considerable.

mainly for elementary occupations such as data recorders, labellers, customer service, farmers, machine operators and for service and sales occupations. Regarding this last point, the main concerns of citizens are related to reconversion and gender issues. Regarding the first, although lower-skilled occupations are not necessarily those that are exposed to a greater risk of automation, they may have significant retraining difficulties.

Along these lines, we will strengthen the Labor Observatory together with initiatives such as Destination Employment, which collect relevant data for the design of public policies that allow us to anticipate market changes, provide recommendations to people for their job search, and guide training policies in view of the new occupations that will be generated. We will carry out this work in

close collaboration with the private sector which, from its reality, can inform on the requirements and possibilities of implementing various policies. These actions will provide information on the impact of AI considering the local reality and new variables, such as company size, economic sector, level of AI adoption, and associated technologies, among others.

· Promote the creation of employment enhanced by technology and support the transition of workers impacted by automation.

The change in the composition of the labor market as a consequence of the emergence of new technologies can have important consequences for those workers who fail to adapt quickly and effectively.

Furthermore, the evidence indicates that people face reconversion in a
ineffective mainly due to lack of information. Along these lines, studies have documented how people, faced with the loss of their jobs, tend to move from a high-risk sector to others with the same characteristics (Oxford Economics, 2019), or are ineffectively trained and retrained. , without taking advantage of the skills they possess.

Consequently, together with the productive sector we will provide more and better information for decision-making in order to recommend work transitions that minimize personal and individual costs. Likewise, we will promote the creation of support instruments so that families can sustain transition periods adequately.

OBJECTIVE 3.2.2.

Provide support to workers in the face of automation.

· Promote critical reflection in around human-machine interaction in the world of work.

One of the most effective ways to facilitate the transition of workers whose functions have been automated or strongly modified is through the generation of intersectoral and multi-stakeholder discussion instances regarding the implications of these processes in the labor sphere. The above is not constituted

It is seen as a count of impacts, but rather, as a process of recognition regarding the expectations, concerns and perceptions of workers.

To do this, we will have mechanisms that encourage reflection on the integration of these systems in a context of reconnaissance.

work version and new relational patterns within organizations. This will allow, when developing and piloting good practice guides, to be based on what was declared by the affected workforce.

· Analysis and prospection of labor regulation with respect to AI and automation.

Based on the discussion and critical analysis of the impacts and transformations of the world of work, we will study the institutionality and labor regulation; We will determine to what extent it can respond to the demands of AI; and we will generate modifications and new bills that point in the direction of people's well-being.

3.3. CONSUMPTION RELATIONSHIPS

The massification of digital commerce (in English) e-commerce has been accompanied by the growing use of AI in consumer interactions between suppliers and consumers. Through the use of AI systems, providers can virtually assist consumers, predict tastes, concerns and preferences of their customers based on their browsing experiences and, accordingly, offer them personalized services. or formulate automated decisions.

However, due to the asymmetry that can characterize consumer relations, the potential of AI is not without risks.

harms to consumer rights, including the generation of unjust or arbitrarily discriminatory results; the absence of transparency in the conditions that make the contracting of products and services possible; or the improper processing of consumers' personal data.

Consequently, providers of products and services that use this technology must ensure that these risks are mitigated.
Additionally, effective mitigation actions should not only be aimed at reducing the risk of harm to consumers, but also at generating trust among the latter regarding the application of AI in consumer relations.

OBJECTIVE 3.3.1.

Promote the use of AI in digital commerce that is transparent, non-discriminatory and respects personal data protection regulations.

· Develop a food ecosystem digital environment conducive to the good use of AI systems that interact with consumers.

We will promote the promotion of ethical and responsible use of AI in consumer relationships, which include but are not limited to automated contracting, liability standards for services associated with AI, among others, so that AI be implemented within the framework of transparent digital commerce, which ensures fair treatment of consumers, in which consumer protection can be adequately monitored.

This involves informing consumers, among other things, about the use of this type of systems; the purpose of the processing of your personal data; and the process by which the provider arrives at a decision that impacts the consumer, for example, by giving or not giving rise to the contracting of a service, or by formulating a personalized offer for a group of consumers.

As a central axis of consumer protection is to seek the formation of informed consent, it is necessary that we ensure that consumers understand, in a timely manner, the implications of the use of AI. For this, the information

tion must be communicated clearly and simple, using understandable language for a consumer without prior knowledge of information technologies.

In order to ensure fair and non-discriminatory treatment of consumers in the context of automated decisions, we will work together with suppliers to ensure the use of explainable and fair algorithms.
These algorithms will be configured with internal review structures, which prevent automated decisions that do not comply with the protection parameters.

of the consumer, in accordance with national legislation. Likewise, these algorithms will enable actions to remedy them, if they do not comply with the legislation.

current. The objective of this type of structures is to prevent systems from operating using discriminatory criteria, such as differentiating contracting conditions or prices according to gender, religion or another.
Understanding that the concept of explainability of AI systems is still under development, we will work to develop standards and good practices in a flexible manner and in accordance with the development and understanding of the operation of this technology.

In short, the promotion of transparent, non-discriminatory trade that respects consumer rights will allow us to fully take advantage of the benefits and opportunities that AI offers with respect to consumer relations.

3.4.

CREATION, INTELLECTUAL PROPERTY AND INDUSTRIAL PROPERTY

As a general-purpose technology, AI has strongly permeated the field of Intellectual Property and Industrial Property (IP). Already between 2013 and 2016, patents in the area had experienced a significant increase in various sectors of the industry (134% in transportation, 84% in telecommunications, 40% in medical sciences and 36% in devices. personal) (World Intellectual Property Organization, 2019) and by 2020 more than 130,000 patent applications have been filed including techniques such as deep learning and neural networks.

The development and adoption of AI have not only opened previously unthinkable questions among experts, such as whether or not a machine can be the owner of intellectual or industrial property rights, but today they also extend to other matters as well. relevant. In a digitalized world scenario, the protection of computer software or programs, data and databases for training AI systems, the definition of patentable subjects, among other topics, seem increasingly essential definitions. .

However, the World Intellectual Property Organization (WIPO) and the various jurisdictions have taken cautious steps and this is currently an ongoing dialogue. For example, there seems to be a consensus on ruling out the possibility that machines may be holders of IP rights, but when it comes to patentable subjects, countries have addressed the protection (or non-protection) of AI systems. by different means. While some have developed patentability criteria through jurisprudence, others have chosen to formalize them in guidelines, discarding or accepting specific cases. In Chile, the criterion must be adopted through a legal and/or regulatory standard.

On the other hand, there are data and databases. They are a raw material of AI systems, so the definition of their protection also stands out as one of the first order. In this regard, in accordance with Law No. 17,336, Chile has a database protection system, through copyright.

Likewise, the value of AI has been highlighted as a useful tool for IP agencies to provide or not protect different asset classes. However, once again there are open discussions regarding

How will its incorporation affect, for example, patentability examinations, given that integrated data systems will contribute to an increase in global, scientific, technological and commercial knowledge.

Even though our country does not appear among the first international rankings in the field of intellectual property, the opening of our borders and the fourth industrial revolution demand that we become part of this dialogue. In it, it will be necessary to consider and make visible possible repercussions of this PI-AI link on the generation and local adoption of AI and address the issue with a renewed and forward-looking view, capable of preserving the appropriate balance between

progress, the rights of creators and society to access knowledge, also providing legal certainty.

Furthermore, the discussion will take place in light of the challenge posed by the integration of AI into complex socio-technical systems. As has been indicated “in the field of inventions

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(Sánchez García, 2018, 52).

OBJECTIVE 3.4.1.

Promote an updated IP system, capable of promoting and strengthening creativity and innovation based on AI, rewarding creators and innovators in a way that encourages them to make their creation and innovation public so that society as a whole can benefit from it.

· Actively participate in dialogue and decision-making bodies on the regulation of AI in the international field of Intellectual Property, maintaining updated information

available to the public on national and international advances.

The IP system currently has a

harmonized international architecture, with standards determined in the Agreement

on Trade-Related Intellectual Property Rights, TRIPS, of the WTO. Furthermore, national IP systems are to this day influenced by

two by international treaties born in the 19th century, the validity of their provisions being reinforced by other multilateral and bilateral agreements.

Given the above, adapting our standards to disruptive technologies such as AI represents a new challenge that must be present when proposing new solutions in the field of IP.

so that they are able to last over time.

For this reason, multilateral organizations such as the WIPO and the OECD that today discuss the relationship between AI and IP, and countries such as Australia, the United States, France, the United Kingdom and the European Union have opened public consultation processes. We will join these and other multilateral bodies, with the aim of making our

state of the situation and be a contribution with the considerations set out in this section, and, in the face of proposals for regulatory changes, we will join the trend of non-binding public consultations.

· Promote the development and adoption of AI, protecting the rights of creators and innovators.

Based on current legislation, by promoting mechanisms that drive the development and adoption of AI, we will maintain the appropriate balance between the rights and obligations of creators and innovators and society as the beneficiary of these creations and innovations.

This represents important challenges in terms of promoting the openness of databases and software, defining patentability criteria for AI systems, as well as developing licenses appropriate to the

internal regulations of our country, but connected with international legal and commercial transit.

· Identify the impact of the standard-tive of IP in AI.

So far, it is unknown whether national IP regulations help or harm the development and adoption of AI, as well as other emerging technologies associated with it, such as cloud computing,

IoT and decentralized protocols based on blockchain

We will carry out registers of inventions and works created for, by or with the assistance of AI systems. We will also generate prospective regulatory analyzes that consider adoption and maturity rates of this technology, addressing patents, copyrights, especially on software and databases, and trade secrets.

· Foster R&D+i in AI for industry creative choice

AI has a high potential to impact the cultural ecosystem and the associated value chain. The implementation of AI can improve the decision-making of the State and creators and the experience of publics and audiences, given that the new

Technologies affect how we access and participate in arts and culture. Therefore, we will promote critical reflection on the uses, limitations and dilemmas that AI poses for the artistic, cultural and heritage spheres.

From this, we will promote the creation of training and education spaces, collaborating with intersectoral AI development platforms for its integration into artistic creation fields and creative economies. In addition, we will promote improvements in data quality and interoperability in all activities associated with

the creative industry, allowing the effective inclusion of AI in this area.

3.5. CYBERSECURITY AND CYBERDEFENSE

In an increasingly digitalized and automated world, cybersecurity has acquired special relevance. The significant increase and complexity of cyberattacks carried out daily adds to the diverse purposes and interests they pursue, as well as the multiplicity of gaps, vulnerabilities and attack vectors. A cyberattack can be as effective and damaging as an armed attack, and even more so in the face of possible warlike uses of these automated systems.

Added to this is the capacity for the propagation of cyberattacks and the potential harm to certain infrastructures or activities, including those that are interdependent, and geopolitical implications, such as

They clarify the application of the principles of International Law on the traditional domains of land, air, sea, space and cyberspace. This undoubtedly implies a current challenge and risk. In fact, the World Economic Forum's 15th Global Risk Report of 2020 places ci-

beattacks within the most worrying scenarios at a global level.

In this context, the progressive adoption of AI and other emerging technologies is also increasing risk factors.

This requires applying control strategies on emerging risks, taking into account the intrinsic limitations of said processes, as well as analysis of new risks and vulnerabilities. So-

Likewise, vulnerabilities in AI systems could produce undesirable effects and lead to incorrect actions. Therefore, adequate risk management requires a dialogue that incorporates the public and private sectors, academia and civil society.

AI is a differentiating element to remain at the forefront in terms of the use of technologies, procedures, equipment and capabilities of national cyber defense, both in legitimate defense actions, as well as deterrence effects and crisis management. In fact, even as AI creates new opportunities for cyber attackers, this technology also has the potential to improve the speed, precision and impact of operational defense, as well as support operational resilience (World Economic Forum, 2020).

Thus, AI is presented as a new tool to keep cyberspace free, open, safe and resilient and, thereby, meet the objectives indicated in our current National Cybersecurity Policy (Government of Chile, 2017). Hence the link between both Policies. AI, in general, can contribute by optimizing response times, identifying vulnerabilities, lities, the detection of intrusions, fraud or identification of in addition to identifying trends malware and/or developing rankings of relevant risks on the network and analyzing large volumes of contextual information while reducing human intervention.

The source and veracity of the information is also a key element for the functioning of these automated systems.
two, cybersecurity being essential for the processes of data collection, storage, processing and transmission. In this sense, the principle of data integrity security becomes of the greatest relevance.

From the above follows the need for us to approach the issue from diverse perspectives, including such relevant matters as the protection in the processing of personal data and the possible compromise and protection of the critical infrastructure of information, services and essential activities. -les of the country. In the same sense, organizations will also require specialized and trained teams in both defensive and offensive aspects, with clearly determined responsibilities and roles.

OBJECTIVE 3.5.1.

Position AI as a relevant component in the field of cybersecurity and cyberdefense, promoting safe technological systems.

· Incorporate AI into cybersecurity and cyberdefense strategies, as well as into bills associated with them.

We will incorporate AI as a component in the next updates of the National Cybersecurity Policy and thus also in cyber defense policies through multi-actor dialogues.

In addition, we will analyze the current legal framework regarding the implications of the use of AI in the field of cybersecurity, and we will give impetus to relevant bills in this regard, such as the Cybersecurity Framework Law Project and Critical infrastructure and the Computer Crime Bill.

· Promote the use of AI systems to react to information attacks tics in the State.

We will generate recommendations for the use of AI in State systems to combat computer attacks. We will do this together with the actors with competencies in cybersecurity matters, to take advantage of the potential of technology.

· Promote training in areas associated with cybersecurity.

We will incorporate training plans on AI for professionals who develop, implement and are in charge of computer systems in the Cybersecurity and Cyberdefense Policies.

Additionally, as progress is constant in both the sophistication of the attacks and defensive techniques, greater research, development and innovation will be promoted in future technologies, for example, quantum technologies, which will provide better tools to address these defensive techniques and training of experts and professionals.

· Incorporate AI into public cybersecurity institutions.

Supreme Decree No. 533, of 2015, of the Ministry of the Interior and Public Security, created the Interministerial Committee on Cybersecurity (“CICS”), with the mission of preparing the country's National Cybersecurity Policy.

We will update this Committee so that it is positioned as an intersectoral space conducive to promoting and carrying out actions in the field of cybersecurity that include AI.

3.6.
GENDER

In the research, technology and innovation system there are important gender gaps. This occurs both at a horizontal level (for example, areas such as engineering or mathematics, which are strongly masculinized, and others such as education or services, which are markedly feminized) and at a vertical level (for example, eg, women excluded from positions of power or decision-making).

Therefore, it is imperative to embrace gender equality as a guiding principle in the policies promoted, that is, that there be no arbitrary discrimination, direct or indirect, against women. In that direction, in January 2020 MinCiencia established a

Gender Roadmap to advance towards the full participation of women in the generation, development and application of knowledge in Chile.

On the one hand, the advantages and improvements that this technology provides can be an efficient engine for overcoming the gender gaps that we have as a society. On the other hand, many of the potential negative effects of AI have their origin in existing biases. in the data with which the algorithms and/or the composition are trained

of the development teams software hardware and related to AI. For example, the World Economic Forum reveals that 78% of AI professionals are men (World Economic Forum, 2018).

In the case of our country, only 1 in 4 enrollments in STEM careers correspond to women (Ministry of Women and Gender Equality, 2018). Practical examples of the effects of this can be seen in facial recognition algorithms that have more difficulty recognizing people.

you are more than men; or in translators who, when moving from a neutral language to one that is not, take preferences when determining the type of a profession.

Finally, the way to address the challenge of gender equity is complex and involves actions in various areas.

considering the complete life cycle of AI systems. Thus, we must work towards more egalitarian development teams, databases free of unwanted bias, audit systems at different stages, among others. The incorporation of women must be as active participants in the creation and implementation, and not as passive beneficiaries.

OBJECTIVE 3.6.1.

Promote the participation of women in areas of research and development related to AI until reaching a level equal to or higher than the OECD.

· Generate a monitoring and analysis system with indicators regarding participation by gender in the research, technology, and innovation system that will include a focus on areas related to AI.

Gender indicators serve to capture the intensity and magnitude of the gaps. We will form a monitoring and analysis system that will contribute to the control of those indicators to maintain an equitable participation rate in areas related to AI.

· Actively promote the access, participation and equal development of women in areas related to AI.

When there are imbalances within the different spheres in which the IA, it is necessary to regularize through affirmative actions such as quotas or criteria tiebreaker in order to reduce inequalities. In this case, women are underrepresented in both access, participation and development, because their capabilities are relegated, especially in areas such as technology. Therefore, affirmative measures encourage finding a fair and ethical balance between those people who are disadvantaged compared to groups that have formed a favored position over time. We will promote these actions that would reform the current balance, since women could achieve those opportunities that would place them on equal terms.

OBJECTIVE 3.6.2.

Promote the participation of women in AI areas in the industry until reaching, at least, a value equal to or greater than the OECD average and ensure that the impact of automation does not harm gender and that job creation is equitable.

· Generate indicators to establish baselines and monitor the inclusion of women in areas related to AI in the industry.

We will develop indicators that allow us to evaluate and compare the gender gap within different related areas
to AI in industry.

For this, we will carry out periodic monitoring of the inclusion of women through observation and early detection of sociocultural patterns that relegate women.

· Incorporate gender variables in the prospection exercises of the
Labor Market, focusing training and updating policies on women to mitigate the impact of automation in areas where They are mostly harmed.

Going deeper into the impacts of automation addressed in the impact on work section, there are sectors, such as manufacturing, that are usually intensive in female labor. The creation of variables and indicators

that determine risk, differentiating by gender, should allow for alerting and reacting in time to accompany the transition and prospect career paths.

Along these lines, we will design and implement training and updating policies with special consideration for women. By segmenting risk by gender, we will design measures with greater focus and effectiveness when mitigating the impact of automation on people's well-being.

OBJECTIVE 3.6.3.

Promote gender equity in the implementation of AI systems.

· Promote gender equity within good practices in the development of AI systems.

The representation of traditionally excluded groups becomes pertinent to protect their interests. Having diverse teams allows for adequate alignment of perspectives, ensuring that the needs of each group find socially and culturally fair solutions. Equity explains how intersectionality affects in the solutions that women could access, since belonging to multiple social categories usually defines the quality of the agreement.

Along the same lines, databases are the raw material for creating algorithms.
Just as they are developed through research, this can be biased in its construction, reinforcing discrimination and stereotypes. Therefore, we will create a guide

of recommendations that, for example, incorporate having databases with a large number of disaggregations to reduce gaps. However, this can generate risks to privacy, so it must be developed in accordance with what is stated in the ethics section.

The relevance of the work with the databases lies in an adequate incorporation of gender, equity and socioeconomic criteria to adequately train the algorithms and will be accompanied by a public story to demonstrate the contribution of women in the disciplines. associated with AI.

· Establish evaluation requirements throughout the entire cycle or life of AI systems to avoid gender discrimination. These criteria must be consistent with the criticality of the application.

Going deeper into what was raised in the ethics section, it is necessary that we carry out a permanent evaluation of the decision-making that the algorithms are making and whether their results do not hide gaps or discrimination.

This preventive measure aims to check if the decisions are biased towards any group, to stop or reevaluate the algorithmic design and give rise to its responsible use.

We will incorporate these considerations in the formulation of projects that involve AI in critical applications in the public sector and will generate recommendations for the private sector as a result of a multi-actor dialogue.

GLOSSARY

5g:
It is the fifth generation of mobile devices and networks, which is characterized by being a high-speed wireless technology, high capacity and low latency (network response time).

Algorithm:
Ordered and finite set of operations that allows finding the solution to a problem.

Anonymization:
Express data relating to entities or people, eliminating the reference to their identity.

Automation:
Application of machines or automatic procedures in carrying out a process or an industry.

Big Data:
These are groups of data so complex and large that to treat them adequately they require non-traditional computer processes and applications.

Blockchain:
It is a data structure made up of blocks. Each one of them (except for the first) is joined to the previous one, forming a line. New data can be added forming a block, joining the last block of the structure.

This structure has been useful for the creation of decentralized protocols that regulate complex interactions in our society (as is the case with currencies, but it can be extended to other activities and tasks).

High-performance computing: Using computing power to solve complex problems in engineering, management, and science. To do this, it relies on certain computational technologies such as supercomputers or parallel computing.

Given:
In the context of AI, they correspond to an observation, representation or sampling of real phenomena, which function as raw material for algorithms, and which allows training and improving algorithms and models aiming to explain or predict the phenomena. re-presented or sampled or observed.

Data center:
Space where the resources necessary for the processing of an organization's information are concentrated. Among its main characteristics are the storage, distribution and processing of data.*

Deep learning: It consists of a Machine Learning learning method, through which an artificial neural network composed of a number of hierarchical levels is used through which information advances to higher levels. This information is processed at each level and once transformed, it goes to the next level, and so on.

Encryption:
Transcription with a key.

Explainability: This term refers to methods and techniques that are used so that artificial intelligence and its results can be explainable, or rather, understood by experts.

Governance:

Art or way of governing that aims to achieve lasting economic, social and institutional development, promoting a healthy balance between the State, civil society and the economic market.

Optical fiber: Thread or bundle of highly transparent glass threads through which information is transmitted over long distances using light signals.

Artificial Intelligence (AI): A computer system that can, for a given set of human-defined objectives, make predictions and recommendations or make decisions that influence real or virtual environments.

Intersectionality:

Intersectionality recognizes that there is no single axis in which discrimination operates, or equity is scarce, or inequalities abound. Thus, it allows us to identify and mitigate oppression or bias in gender, with respect to income, representation, participation, ethnicity, sexual orientation, or other categories.

IoT or Internet of Things: A digital interconnection of everyday objects with the Internet.

Machine learning: Scientific discipline of artificial intelligence that designs and builds systems that learn automatically by identifying complex patterns in a universe with large amounts of data.

Malware:

Any type of software that performs harmful actions on a computer system for intentionally and without consent.

Equitable participation: Participation through the promotion of more widespread access to economic, social and political opportunities.

Computational thinking: It is the ability of an individual to analyze and solve problems using acquired skills that are typical of computing and critical thinking, among others.

Job reconversion:

Process through which it is proposed to adapt people to new work configurations, such as, for example, new technologies.

Reskilling: Training workers in a new skill in response to technological change in the organization. In other words, it involves “professional retraining” of workers.

Sandbox:

Process isolation system or isolated environment, often used as a safety measure.

Bias:

Systematic error that can be incurred when, when sampling or testing, certain responses are selected or favored over to others.

Software:

A set of programs and routines that allow the computer to perform certain tasks.

STEM:

Acronym for the English terms Scien-ce, Technology, Engineering and Mathematics.

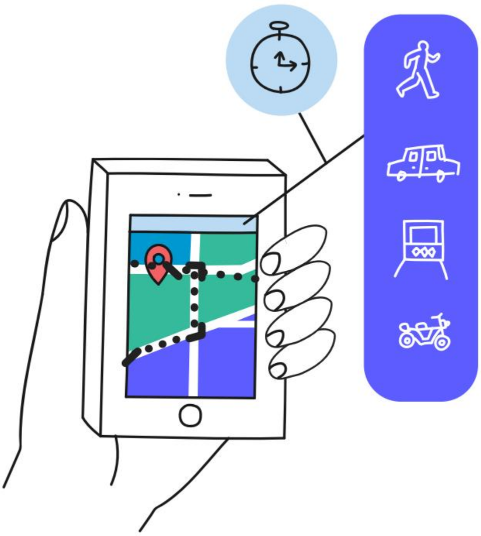
Digital transformation: Changes associated with the application of digital technologies in all areas of daily life and society.

Cloud technology (cloud computing): Technology that allows computing services to be offered over a network, which is generally the Internet.

General purpose technology: Corresponds to a technology that has the potential to impact the social and economic structures of the entire society.

Upskilling:
Additional training that workers receive, which helps them improve their personal skills and thus perform their job tasks more efficiently.

Last mile: Final stretch of a communication line, whether telephone or optical cable, that reaches the end user.



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